



Technical Letter Report
TLR-RES/DE/REB-2025-02

Evaluating the high temperature design rules and design space of Section III, Division 5 and Section VIII of the ASME Boiler & Pressure Vessel Code

Date:

March 2025

Prepared in response to Task 6 in User Need Request NRR-2022-009, by:

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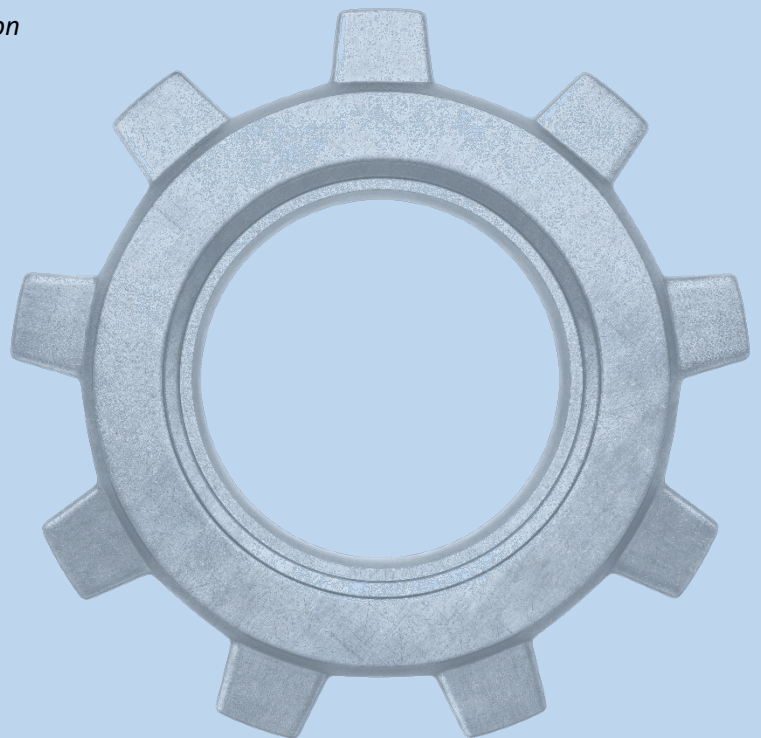
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Evaluating the high temperature design rules and design space of Section III, Division 5 and Section VIII of the ASME Boiler & Pressure Vessel Code

Applied Materials Division

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March 2025

EXECUTIVE SUMMARY

This report considers the applicability of the American Society of Mechanical Engineers (ASME) Boiler & Pressure Vessel Code (BPVC) Section VIII, “Rules for Construction of Pressure Vessels,” for the design and construction of components for high temperature nuclear reactors. While the Nuclear Regulatory Commission (NRC) endorses use of ASME BPVC Section III, “Rules for Construction of Nuclear Facility Components,” Division 5, “High Temperature Reactors,” some non-light water reactor vendors have expressed interest in applying Section VIII.

This report analyzes the differences between the high temperature design rules in Section III, Division 5, focusing on Class A construction, and the Section VIII rules, including both Divisions 1 and 2. For Section VIII, Division 2, both design by analysis and design by rule are considered. The report addresses both the differences in design methods and design data. In addition, the report compares the feasible design space implied by the different sets of rules by evaluating a simple, but representative, parametric design problem. These analyses demonstrate the differences in design approaches and qualitatively compares the design rules for several key variables. These analyses showed that Section III, Division 5 rules generally allowed for lower primary and secondary stresses than the Section VIII rules. This is particularly true at longer service times and higher temperatures.

There are limitations and advantages for both Section III and Section VIII depending on the needs of the specific application. These Codes provide rules on multiple aspects of high temperature construction including quality assurance, design, fabrication, installation, pressure protection, etc. This report primarily focuses on design considerations. Some of the key design differences between the Section III and Section VIII rules included: 1) design life philosophy, where Section VIII applies an indefinite life concept (time-independent allowable stresses) while Section III, Division 5 includes time-dependent allowable stresses; 2) the definition of the allowable stresses, even at a fixed time; 3) addressing failure modes, particularly high temperature cyclic effects such as creep-fatigue; 4) welding rules; 5) which materials can be used; and 6) the type of components the codes are applicable to. While these differences are important, it should be noted that components which are designed to the Section VIII rules often utilize a fitness for service program to provide information on the remaining life of a component. While these programs are not explicitly part of the Section VIII requirements, historically, they have been used to help ensure component reliability in non-nuclear industries.

An in-depth discussion of the differences between Section III, Division 5 and Section VIII is provided in Chapter 3 of the report. With these differences identified, this report details topics which may need consideration to make the Section VIII rules applicable to high temperature nuclear construction. Some considerations are proposed to allow for the development of screening criteria to extend the applicability of Section VIII to high temperature nuclear applications. Examples include the application of a negligible creep criteria and fatigue criteria.

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1 Introduction

1.1 Overview of the report

This report provides a discussion on the potential to use components designed and constructed to the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code (BPVC) Section VIII, “Rules for Construction of Pressure Vessels,” in high temperature reactors. A note on the title: this report seeks to compare the design rules and the design space across the Section VIII and Section III, “Rules for Construction of Nuclear Facility Components,” Division 5, “High Temperature Reactors,” Codes. In this report, design rules refer to the procedures and practices used in each section of the Code to develop a component design. Additionally, design space means the range of design variables or parameters that can pass all the corresponding design rules, including the component materials, temperature, and loading conditions.

This report is limited to the design methods in Section III, Division 5 and Section VIII. There are differences in fabrication, pressure testing, pressure relief, and pre-service inspection requirements between the Section VIII and Section III, Division 5 rules. These topics are not addressed in this report.

This report specifically considers the 2023 editions of Section III and Section VIII, the most recent editions at the time of writing. New Code editions are published on a two-year cycle. While it is anticipated that much of the discussion in the report will hold true for future editions, a reader should consider relevant Code updates.

The report, as noted in the title, focuses on the Section III, Division 5 rules, which are for high temperature reactor components. The review then focuses only on the parts of Section VIII that allow for design at high temperatures. For design by rule in Section VIII, Division 1 and Section VIII, Division 2, Part 4, this work does not make detailed comparisons between the design rules for a wide set of component types. The design by rule formula would need to be checked against detailed design by analysis calculations with detailed finite element analysis models of the corresponding component types. The effort required for these calculations is out of the scope of this project.

Note there are differences in how Section VIII and Section III define high temperature (the start of the creep regime). For Section III the transition is a fixed temperature, set to the same value for a broad category of materials. For Section VIII, the creep regime starts at the temperature at which the values of the allowable stresses in Section II, Part D are italicized, indicating they are controlled by time-dependent data. This means that there is some overlap between Section VIII, Division 2, Class 1 construction and the creep regime as defined in Section III. However, for simplicity, this report does not consider this narrow, material-specific range of temperatures.

The following chapters of the report are laid out in the following way. Chapter 2 provides a brief overview and primer on ASME design methods in general. Chapter 3 summarizes the high-level design space differences between the Section III, Division 5, Subsection HB, Subpart B design rules and the design rules in Section VIII, Division 1 and Division 2. Chapter 4 provides a means to analytically compare the design rules across a wide range of design space, including different primary and secondary loads, temperatures, design lives, and load cycle types. While this analysis must, by necessity, greatly simplify the design problem to cover such a broad design space, this analysis is informative in that it clearly demonstrates the differences between the sets of design rules. The analysis is also potentially quantitatively useful in evaluating the feasibility of applying the Section VIII design rules, as results generated from an analysis of the component selected for this study, the Bree cylinder, have been widely generalized to other components in the past. Finally, Chapter 5 summarizes this report and briefly outlines strategies for mitigating the differences between the sets of design rules summarized here.

1.2 Overview of Section III, Division 5

The Nuclear Regulatory Commission (NRC) has endorsed Section III, Division 5 for construction of non-light water reactors in Regulatory Guide 1.87 Revision 2 [1]. Draft Guide 1436 has been published and contains the proposed update to Regulatory Guide 1.87 and endorsement of the 2023 edition of the Code [2]. For Class A components, Section III, Division 5 is a general design by analysis code, allowing the design of arbitrary components including pumps, valves, and piping. Section III, Division 5 contains an additional set of design and construction rules in Subsection HC for Class B components. Currently, these rules adopt a design by rule approach that, at least approximately, mimics Section VIII, Division 1. Current work by the cognizant Code Committees aims to expand these rules to include a simplified design by analysis option that covers the key Class A design limits, including creep-fatigue. The Class A rules are the focus of this report when discussing Section III, Division 5 and the Class B rules are generally not discussed.

Currently, Section III, Division 5 provides only six materials for Class A construction:

1. 304H stainless steel
2. 316H stainless steel
3. Alloy 800H
4. Grade 91 steel (Type 2)
5. Grade 22 steel in the annealed condition
6. Alloy 617

One of the primary reasons for the limited number of materials included in Class A is the substantial amount of material data needed for qualification.

As Section III, Division 5 was developed for high temperature nuclear applications it addresses many of the anticipated high temperature failure modes. Class A components guard against six failure modes – 1) time-independent plastic instability, 2) time-dependent creep rupture, 3) creep-fatigue damage, 4) time-dependent cyclic excessive deformation, i.e. ratcheting, 5) time-independent buckling, and 6) time-dependent buckling. Additionally, commensurate with the consequence of failure of nuclear components, there is substantial conservatism built into the design rules.

1.3 Overview of Section VIII

Section VIII, like Section III, Division 5 covers the design of components operating at elevated temperature, but Section VIII is used by non-nuclear industries, for example in petrochemical applications. Section VIII covers the design and construction of vessels though broadly construed to include components like heat exchangers and connections, including bellows. Section VIII includes Division 1 (VIII-1), which is an exclusively design by rule approach, and Section VIII, Division 2 (VIII-2), which covers both design by rule in Part 4 and design by analysis in Part 5.

There are more materials available for use with Section VIII construction than there are for construction to the Section III, Division 5 rules. Section VIII makes hundreds of ferritic and non-ferritic materials available to the designer. This larger material library provides more design flexibility, either to accommodate specific environmental concerns or component-specific criteria.

Section VIII rules do not consider several key design limits and material degradation mechanisms that are considered by Section III, Division 5. Specifically, as discussed further in this report:

1. Section VIII adopts an indefinite life approach for allowable stresses, where the material allowable stress remains constant regardless of the expected or actual service time of the component. By contrast, Section III, Division 5 rules allow for changes in the allowable stresses, due to creep damage thereby impacting the component design life.
2. The joint efficiency factors applied to the allowable stress for welded joints in Section VIII do not account for the reduction in creep strength (even for defect-free welds) of a weldment compared to base material.
3. Some of the Section VIII rules consider pure fatigue damage but do not cover creep-fatigue interaction as a potential failure mode.

This report addresses high temperature design. One consequence of this decision is that this report only considers Section VIII, Division 2, Class 2 construction. Class 1 vessels are designed to the allowable stresses given in Section II, Part D, Table 2 while Class 2 vessels are designed to the allowable stresses given in Section II, Part D, Table 5 – which are generally greater than the corresponding allowable stresses in Table 2. In general, this means a Class 2 vessel will use less material, than a Class 1 vessel, as the Class 2 vessel wall can be thinner, based on the larger allowable stresses. There are also differences in the quality requirements between the two classes, which are not discussed here. Additionally, design by analysis is always allowed for Class 2 (and in fact required if no design rules exist), whereas design by analysis is only allowed for Class 1 in the absence of a design by rule formula.

More relevant for this report, the allowable stresses in Table 2 do not extend into the creep regime, per the Section VIII definition and Section VIII, Division 2, 5.1.1.3.1 restricts Class 1 construction to below the elevated temperature regime (with exceptions if deemed acceptable to the Authorized Inspector). As such, this report only considers the Class 2 design rules in this report. Moreover, Section VIII, Part 2, 5.1.1.3 requires the use of the design by analysis rules to the elastic methods (when the fatigue screening criteria are satisfied) for components operating in the time-dependent regime.¹ We limit the remainder of the report to consider only the elastic methods.

¹ There appears to be a gap in the Section VIII, Division 2 rules in the provisions allowing the use of the elastic design by analysis rules for elevated temperature design. 5.1.1.3 allows the use of the elastic stress analysis procedures in 5.2.2, 5.3.2, 5.6, 5.7.1, 5.7.2, and 5.8 provided the fatigue screening criteria of 5.5.2.2 are satisfied. This list does not include the ratcheting assessment rules in 5.5.6. However, 5.5.1.6 requires a ratcheting evaluation even if the fatigue screening criteria are met. This means that, for components operating at elevated temperatures, even if the component passes the screening criteria the designer has no way to meet the required ratcheting evaluation. The authors suspect this is an oversight in the Code rules, rather than the intent of the Code.

2 Background: A primer on ASME Design Rules

This report assumes a basic familiarity with ASME design methods. This section gives a brief primer on common methods and concepts used widely throughout the ASME Code, including in both Sections III and VIII. This section only provides a brief, high-level overview. For additional information the reader can consult references [3] and [4].

The ASME design methods generally consider the component under the loads in two conditions: the steady state, where the loads are held constant for an infinite amount of time, and the cyclic steady state, where the loads are cyclic, following some assumed set of periodic design transients, but only the component response is considered in the limit of applying and reapplying the transients many times. Both types of analysis give a different stress and strain distribution. For the steady analysis this will be a single value of stress at each point in the component, for the steady cyclic analysis it will be a single time history of stress and strain covering a single, periodic, steady cycle. The design calculations compare these steady and steady cyclic responses against design limits to determine if the component is acceptable.

A complication is that calculating the actual component steady state response is difficult, and so the Code rules will typically approximate the response from some simpler form of analysis. The classical approach approximates the actual component response with a linear elastic stress analysis, i.e. “design by elastic analysis.” Other design approaches in the Code approximate the steady component response with an elastic, perfectly plastic material model. Finally, in Section III, Division 5, the Code has an option for conducting a full inelastic stress analysis, which directly simulates the component response – subject to the accuracy of the constitutive model used in the design analysis. The remainder of this section, and the report in general, focuses on design by elastic analysis. This is the oldest and most widely used method and still has definite advantages over other approaches because the stresses are quickly and easily calculated and individual load combinations can be superimposed from separate analyses, rather than requiring developing composite cycles that combine individual design transients into a single cycle. Moreover, Section VIII, Division 2, 5.1.1.3 restricts design by analysis for Section VIII in the high temperature regime to the elastic methods only (see above for a footnote discussing one potential caveat or limitation to applying the elastic rules to elevated temperature design).

2.1 Steady analysis

To start with the steady analysis, consider a component operating in the creep regime subjected to typical loads, which might include internal pressure, thermal stresses, and mechanical forces transmitted from other, connected, components. For an elastic analysis a designer could analyze each type of loading individually and superimpose the resulting stresses. Thinking then about each type of load individually and considering the actual, long-term behavior under creep conditions results in two different types of loads.

The first type of load is called primary load in the ASME Code. These are loads, and resulting stresses, that creep and plasticity cannot reduce through stress relaxation. The canonical example is an average membrane stress due to internal or external pressure. The average stresses in the component induced by pressure cannot diminish with time because they are statically necessary to maintain equilibrium with the constant pressure. Average stress here means the reaction generated by the component stresses resisting the pressure – the distribution of the stress can change with time, but the reaction always must equilibrate the pressure.

The second kind of load and resulting internal stress are called secondary in the ASME terminology. The canonical example is a thermal stress. Stress relaxation can reduce these types of component stresses to zero with infinite time, for an idealized power law creep model, and inelastic deformation because they are not statically necessary to oppose an applied, external load. An equivalent definition is that secondary stresses are self-equilibrating: they do not require an external reaction to satisfy equilibrium.

ASME introduces a third category of stress, called the peak stress. These are local stresses caused by stress concentrations that will not affect the overall equilibrium of a component. These stresses might redistribute with time but do not affect net equilibrium of the component with the external loads. A canonical example is the stress near a hole in a vessel for a nozzle. These stresses will be larger than the average stress across the vessel section but will remain localized near the nozzle itself and will not affect the overall equilibrium with the internal vessel pressure. As such, they can be dealt with separately from the primary and secondary stresses if the designer is only concerned with the plastic collapse or creep rupture of the vessel.

ASME primary and secondary stresses are net section stresses – a designer is primarily concerned with their effect on the entire vessel section. And, as noted previously, while the net reaction of a primary stress must remain the same, the stress distribution across the section can change with time. ASME considers two subcategories of stress to deal with the effect of stress redistribution – membrane and bending stress. The membrane stress is the average stress across the section while the bending stress is often taken to be the first moment of the stress distributed across the section. For more complex stress distributions, the remaining higher moment stresses are often deemed to be peak stresses. The process of allocating stresses to these different categories is called “linearization”, for more details see [5].

The reason linearization is needed is stress redistribution. The elastically-calculated membrane stress (for a structure with a single load path or considering a section of a vessel in isolation) must equal to the actual membrane stress accounting for inelasticity. For bending stresses however, the elastically-calculated stress distribution will be different from the fully-plastic stress distribution, which will again be different for the stress distribution under long-term power law creep. The classical way to account for these differences is to use an inelastic section factor giving the ratio between the elastically-calculated maximum stress across the section and the stress under the fully-plastic or steady creep solutions. The ASME Code provides these section factors, but they can only be applied to the bending portion of the loading, hence the need for linearization.

There is no simple way to distinguish primary and secondary stress from an elastic analysis. The Code applies the concept of stress “classification” to classify elastically-calculated stresses into these two categories. There are several, mostly-equivalent ways to think about stress classification:

1. Strictly, stress classification is defined through Code rules through stress classification tables. An example is Table HBB-3217-1 in Section III, Division 5. This table tells the designer how to categorize a wide variety of types of loading.
2. More fundamentally, a primary stress is required for the component to remain in equilibrium with the external loads, while a secondary stress is not. The designer can classify stresses from this fundamental definition.
3. Finally, both stress linearization and classification can be best understood in the historical context of elastic stress analysis for ASME component design.

When the modern Code rules were first formulated in the 1960s, Finite Element Analysis (FEA) was an emerging technique and not in common usage. Instead, axisymmetric bending analysis was the best available technology for design by analysis approaches which required calculating the stresses in the component. Frame bending analysis accounts for membrane stress along the section under consideration as well as bending and shear stresses across the section. Under this framework it is immediately clear how to linearize stresses – the bending and membrane stresses are treated in a distinct fashion in the analysis itself.

A statically determinant frame structure is one where the internal stresses can be calculated directly from the external free body diagram expressing equilibrium with the applied loads. Such structures are not common in vessel design. Instead, most components will be indeterminant, meaning solving for the stresses requires including the elastic stiffness and deflection of the vessel. One way to use frame analysis to solve for the stresses and strains in such a structure is to first artificially break the component at locations to make the component statically determinant, solve for the statistically determinant stresses, and then reimpose the appropriate constraints to calculate a second set of stresses required to deform the “released” component back into the correct configuration.

With this idea in mind, the concepts of primary and secondary stress classification are inherent parts of the stress analysis. The first set of stresses calculated for the statically determinant structure are the primary stresses – by definition this is a set of stresses required to maintain equilibrium with the applied loads. The second set of stresses calculated by bringing the component back into mechanical compatibility are the secondary stresses. Mechanical compatibility means the relaxed constraints in the initial, primary stress analysis are reimposed by applying appropriate stresses, i.e., the secondary stresses. These stresses bring the structure back into appropriate alignment – at a minimum displacement continuity and, depending on the type of connection, rotational continuity as well. The frame analysis approach could not account for stresses near local discontinuities like nozzles, so instead these stresses were calculated based on the membrane stresses and a, often experimentally-determined, stress concentration factor. These are then the peak stresses.

Even though stress analysis is now almost exclusively done with FEA, this historical overview clearly shows (1) where the concept of linearization and classification came from and (2) illustrates two well-known points of interest in the general approach, specifically:

1. The primary and secondary stresses are not unique. There are often many ways to release a frame model to make it statically determinant, which will result in many different distributions into primary and secondary stress.
2. While a thermal stress is the canonical example of a secondary stress, stresses resulting from pressure loads will also be partly secondary where breaking the component into pieces is required to complete the statically determinant analysis. These stresses occur at structural discontinuities, like the connection between the shell and head of a vessel.

Additionally, though not immediately clear from this description, overclassifying a secondary stress as a primary stress is conservative when executing the design check to compare the primary stress versus some allowable stress.

This historical context is useful in understanding how the concepts of stress linearization and classification were formulated, but modern practices for performing linearization and classification are different now that FEA is often used for the stress analysis. Typically, designers will run two elastic stress calculations, one with only the pressure and other mechanical stresses and the second with only the thermal stresses. All stresses in the first analysis are considered to be primary and all the stresses in the second analysis are considered to be secondary. This approach overclassifies some secondary stresses, present in the first FEA analysis, as primary, but that is a conservative approximation.

Next, the designer selects several Stress Classification Lines (SCLs) for analysis. The Code rules require passing the allowable stress checks at all locations in the vessel, but in practice engineering experience can be used to limit the detailed analysis to a few key SCLs.

For each set of FEA results and for each SCL the design performs linearization following a procedure similar to the ones documented in WRC-429 [5]. This is an automated approach to taking the stresses from the FEA along that line and calculating membrane, bending, and peak stress. Some finite element software can do this automatically. The peak stresses from both sets of analyses can be summed to give the total peak stress, while the individual membrane and bending components in the pressure-only and thermal-only analyses give the primary and secondary membrane and bending stresses.

Regardless of how they are calculated, after formulating the primary membrane and bending stresses, they must be converted to scalar stress measures. Conventionally ASME uses the Tresca (maximum shear) stress, which is a reasonable choice of failure theory for considering plastic collapse of the vessel. The scalar bending stresses can be corrected for stress redistribution with a plastic section factor and then the sum of corrected bending and membrane are compared to an allowable stress. The next chapter discusses the actual formulation of the Section III and Section VIII allowable stresses, but this basic description encompasses how all ASME design methods carry out the primary load limit check.

2.2 Cyclic analysis

The approach to cyclic analysis varies greatly from section to section in the ASME Code. The following sections provide a detailed comparison of the exact methods used in Section III, Division 5, Section VIII, Division 1, and Section VIII, Division 2. However, there are some commonalities across these and the rest of the ASME cyclic design rules.

Generally, the cyclic design rules consider two design limits: excessive strain accumulation and local initiation of fatigue (or creep-fatigue) damage. Generally, the Code considers these limits in the context of the steady cyclic behavior.

Under repeated application of a single load cycle a component, subject to a few reasonable assumptions related to the inelastic response of the component materials, will reach a steady state characterized by a periodic, repeating set of stresses and strain rates [6]. The strains themselves can fall into two general categories: periodic and repeating, called shakedown behavior, or increasing by a steady strain increment each cycle, called ratcheting behavior. At a minimum, a component should be designed to avoid ratcheting as a ratcheting design will continue to accumulate strain until it inevitably exceeds a given strain limit, regardless of what the specific strain limit might be.

Within the “safe” shakedown bounds there are three distinct subsets of behavior. In the first subset the component in the steady state behaves elastically – that is, in the steady stress-strain cycle the deformation is entirely elastic – though prior to reaching the steady state behavior it may accumulate some inelastic strain. This is called elastic shakedown. In the second subset the strains are periodic, meeting the shakedown definition, but the steady cycle involves some inelastic deformation. This is called plastic shakedown. The third subset falls under the elastic shakedown category but is a more specific condition where the component never deforms inelastically, even before it reaches the steady, shakedown state.

Generally, the Code tries to avoid plastic shakedown as the repeated inelasticity in the steady cycle suggests the potential for the material to accumulate fatigue damage. However, this is not universal across all the Code design rules. For example, the Section III, Division 5, Subsection HB, Subpart B rules allow for plastic shakedown, so long as the detailed creep-fatigue damage check passes the Code criteria.

Actually determining which type of cyclic behavior a component will manifest requires a detailed inelastic analysis. The Code uses the concept of a Bree diagram [7] to provide a simplified method for making this determination based on elastic analysis. Chapter 4 describes the derivation of the classical Bree diagram in detail. Briefly, a Bree diagram summarizes the steady cyclic behavior of a particular component in a 2D diagram parameterized by the primary load on one axis and the secondary load on another axis. Even though these diagrams are specific to one particular component geometry, the Code applies them to general types of components with the concept of stress classification discussed previously. The designer can enter the diagram using their classified primary and secondary stresses and determine if the component will shake down or not and which specific shakedown behavior it will experience (elastic, elastic shakedown, or plastic shakedown). This concept can be extended to also provide more detailed estimates of the accumulated strain, for example with the O'Donnell-Porowski method used in Section III, Division 5 [8].

The Code's fatigue rules vary too much to briefly summarize. In general, they require constructing an approximation to the cyclic steady state stresses and strains by manipulating the results of the elastic stress analysis, calculating quantities based on this steady behavior, and using these quantities to evaluate fatigue or creep-fatigue damage. The following chapter discusses the specific Section III and Section VIII methods in greater detail.

3 Summary of design space differences

3.1 A high-level overview of the Section III, Division 5 and Section VIII design rules

ASME design rules follow one of two general approaches:

1. Design by rule, in which the designer does not do detailed stress analysis but rather designs the component geometry in accordance with design formula provided by the Code.
2. Design by analysis, where the designer performs a stress analysis of the component, often with FEA, and determines the acceptability of the design by applying criteria based on those stresses, provided by the Code.

Design rules aim to guard against certain expected failure modes. For design by rule these failure modes may not be explicitly stated in the design rules, though they are often implicitly understood. Design by analysis approaches will typically state these failure modes explicitly. In evaluating the design rules considered here, this report considers five potential failure modes:

1. Plastic collapse
2. Creep rupture
3. Excessive strain accumulation and/or ratcheting
4. Local fatigue or creep-fatigue failure
5. Buckling, either time-independent or time-dependent creep buckling

Design rules can cover low temperature design, high temperature design, or both. The approaches described here cover both high temperature and low temperature design, but the focus of the report is only on high temperatures.

The design rules considered here take two different general approaches to high temperature design. At high temperatures, materials and components cannot be expected to last indefinitely. High temperature degradation mechanisms like creep and creep-fatigue damage will inevitably cause a component to fail in a finite amount of time. One high temperature design approach accounts for the finite life of the component by designing against a design life – the target minimum life of the component in operation. Another approach is to not target a definite life but rather design for notionally indefinite life, while acknowledging through inspection, repair, and remaining life assessment approaches that the component may fail while in service.

This report considers four general sets of design rules, one for Section III, Division 5, and three for Section VIII:

1. Section III, Division 5, Class A, described in Subsection HB, Subpart B of the Code.
2. Section VIII, Division 1.
3. Section VIII, Division 2, Part 4, Class 2.
4. Section VIII, Division 2, Part 5, Class 2.

As noted above, Section VIII, Division 2, Class 1 construction is not considered as it does not provide for elevated temperature design in the definition used in Section VIII.

Table 3.1 categorizes the four sets of design rules by the criteria outlined in this section. While the four methods are generally comparable, there are specific limitations or differences in the design space covered by each of rules. The rest of this chapter delves into these differences.

Table 3.1. General comparison of the four design methods.

	Section III, Division 5, Class A	Section VIII, Division 1	Section VIII, Division 2, Part 4, Class 2	Section VIII, Division 2, Part 5, Class 2
Approach	Design by analysis	Design by rule	Design by rule	Design by analysis
Design life?	Yes	No	No	No
Plastic collapse	Yes (multiple methods, multiple allowable stresses)	Yes (single allowable stress)	Yes (single allowable stress)	Yes (multiple methods /single allowable stress)
Creep rupture	Yes (multiple methods, multiple allowable stresses)	Yes (single allowable stress stress)	Yes (single allowable stress)	Yes (multiple methods /single allowable stress)
Strain accumulation	Yes	No	No	Yes (ratcheting only, no explicit calculation of strain)
Fatigue	Yes	No	Yes (fatigue exemption only)	Yes
Creep-fatigue	Yes	No	No	No
Buckling	Yes (time independent and time dependent)	Yes (time independent)	Yes (time independent)	Yes (time independent)

The reference point for the plastic collapse and creep rupture analysis for the design by analysis methods is the traditional ASME approach to allowable stress design where the stresses on the component are categorized into primary, secondary, or peak stress categories and the primary stresses checked against allowable stress limits. This is an approximate means to account for plastic and creep stress redistribution in a redundant design, i.e. one where inelastic stress redistribution before collapse is possible. Both design by analysis methods provide alternative means to completing the load redistribution analysis based on elastic-plastic analysis.

The “allowable stress” notation in the table indicates that the definition of the relevant ASME allowable stress includes criteria aimed at guarding against that failure mode. That is, the allowable stress “covers” the relevant failure mode. For plastic collapse this includes the material yield and ultimate tensile strength. For creep, this at a minimum includes creep rupture data, but might include additional types of data as well like creep deformation limits. These allowable stresses are one of the main mechanisms in providing safe and effective designs based on material failure data and, for the design by rule approaches, they are the only real mechanism for including material data into the Code rules.

3.2 Design life versus indefinite life philosophies

3.2.1 Definitions

The previous section outlines the difference between a design life versus an indefinite life philosophy. Section III, Division 5 components are designed to meet a target service life (the design life) provided by the Owner, whereas Section VIII components are designed for indefinite life.

When comparing equivalent designs, this means that intrinsically a Section VIII design may lose margin between the allowable and experienced stresses as the component continues to accumulate time in service. This contrasts with a Section III, Division 5, Class A component which will not lose design margin as time goes on (up to the design life).

The Section VIII time-dependent allowable stresses are anchored against extrapolated 100,000-hour properties but this should not be thought of as an inflection point, i.e., consider Section VIII designs adequate for service lives less than 100,000 hours but not for lives greater than 100,000 hours. Several factors should be considered against making such a judgement:

1. The 100,000-hour time is arbitrary, set by committee judgement for safe and adequate designs for indefinite life.
2. The 100,000-hour data has safety factors applied to the material data and, for some criteria, based on lower bound material properties.
3. Detailed fatigue analysis based on the actual component (expected) service life may provide an additional cap on the design of Section VIII components.

Under certain conditions, Section VIII components may continue to operate successfully well beyond 100,000 hours. To manage the risk of components operating beyond the time used in formulating the allowable stresses, plants operating Section VIII components often maintain active remaining life assessment programs. These programs monitor plant components as they age and use that monitoring data to update the expected remaining life of the component in actual operation. Examples of these types of methods can be found in ASME FFS-1/API-579. While such fitness for service programs are, strictly speaking, outside the Code design space, they certainly contribute to the longevity of Section VIII components in service. Additionally, many operating components do experience failures in service but can be successfully and economically repaired and returned to service, again further extending the actual life of the component in service.

3.2.2 Discussion

This difference in design philosophy reflects a difference in how components are expected to operate. A Section VIII component, even properly designed, may not be expected to function the full life of the system. If a component fails, oftentimes it can be subsequently repaired and returned to service but, sometimes, it may need to be removed from service and replaced. For safety or operationally critical components, operators of Section VIII components will often conduct supplementary inspection and remaining life assessment to determine an expected interval until component failure and, therefore, repair or replacement. By contrast, a Section III, Division 5 component is designed in such a way that it is not anticipated to fail for the full component design life.

3.3 Definition of allowable stresses

The four sets of Code rules reference four different allowable stresses. The Section III, Division 5, Class A rules utilize an allowable stress, S_o and an allowable stress, S_{mt} . S_o follows an indefinite life approach but applied to design conditions while S_{mt} follows a design life approach applied to operating conditions. This means a Class A component is conservatively designed against both indefinite life and design life philosophies. In general, the design of components with a shorter design life will be controlled by S_o while components with a longer design life will be controlled by S_{mt} . The definition of the allowable stress S_o matches the definition used in Section VIII, Division 1, though the numerical values provided in the Code in Section III, Division 5 and Section II, Part D have diverged over the years. Likely, this is because the Section II, Part D tables historically were updated more frequently than the Section III, Division 5 tables and the underlying datasets (of tensile and creep data) have diverged over time. This correspondence in the definition reflects the reason the Section III, Division 5 rules apply the S_o check: component with short design lives could pass the time-dependent S_{mt} check but fail the Section VIII rules, where the allowable stresses are based on extrapolated 100,000-hour properties. Including the S_o check screens out these designs, with the goal of ensuring all components that pass the Section III, Division 5 rules will also pass the Section VIII, Division 1 criteria.

The Section VIII design methods reference the allowable stress S contained in Section II, Part D, Table 1 (Division 1) and Section II, Part D, Table 5 (Division 2, Class 2). These allowable stresses all follow an indefinite life approach.

The creep criteria in the Section VIII allowable stress definitions, and S_o for Section III, Division 5, use extrapolated 100,000-hour properties. This does not mean Section VIII components have a 100,000-hour design life; Section VIII applies an indefinite life philosophy. Rather the cognizant Code Committees deem the extrapolated creep data at 100,000 hours to be a reasonable basis to design components against for elevated temperature service with indefinite life.

Table 3.2 summarizes the criteria used in each of the four different allowable stress definitions. For S_{mt} the creep properties are extrapolated to the design life. For the remainder of the allowable stresses based on an indefinite life approach the creep properties are based on extrapolated 100,000-hour data. This section and table only consider base metal; see below for welds. The allowable stress is always the minimum of the values listed in the table for each temperature and time.

Table 3.2. Allowable stress criteria. Section III, Division 5 indicates Section III, Division 5. II-D indicates Section II, Part D.

	Section III, Division 5 S_o	Section III, Division 5 S_{mt}	II-D, Table 1 S	II-D, Table 5 S
Approach	Indefinite	Design life	Indefinite	Indefinite
Tensile	$0.67 \sigma_{y,min}$	$0.67 \sigma_{y,min}$	$0.67 \sigma_{y,min}$	$0.67 \sigma_{y,min}$
	$0.67 \text{ to } 0.9 S_y$	$0.67 \text{ to } 0.9 S_y$	$0.67 \text{ to } 0.9 S_y$	$0.67 \text{ to } 0.9 S_y$
	$0.285 \sigma_{u,min}$	$0.333 \sigma_{u,min}$	$0.285 \sigma_{u,min}$	$0.417 \sigma_{u,min}$
	$0.285 S_u$	$0.333 S_u$	$0.285 S_u$	$0.417 S_u$
Creep	$1.0 \sigma_{minrate}$	$1.0 \sigma_{1\%}$	$1.0 \sigma_{minrate}$	$1.0 \sigma_{minrate}$
	$0.67^2 \sigma_r$	$0.8 S_{tc}$	$0.67^1 \sigma_r$	$0.67^1 \sigma_r$
	$0.80 S_r$	$0.67 S_r$	$0.80 S_r$	$0.80 S_r$
Other	Currently limited to the lowest value of S_{mt} for the given temperature			

² Strictly F_{avg} with $F_{avg} = 0.67$ for temperatures less than or equal to 815° C. For temperatures that exceed 815° C, F_{avg} is determined from the slope of the log time-to-rupture versus stress plot at 100,000 hours so that $\log F_{avg} = 1/n$, but not greater than 0.67, with n the slope.

For the yield stress, the factor on the design stress ranges from 0.67 to 0.9. In general, this factor is set by the Code Committees based on the ductility of the material, with more ductile materials allowed to have higher values. In principle the factor should be the same across all four allowable stresses, so for comparison's sake this report assumes a common value.

In these definitions this report uses σ to represent an average value and S to indicate a statistical lower bound. Conventionally the Code uses a 90% lower confidence interval on that data as a minimum property, though this is not formally specified in the Code.

In this table $\sigma_{y,min}$ is the minimum specified (room temperature) yield strength for the material and $\sigma_{u,min}$ is the minimum specified (room temperature) tensile strength. S_y is the design yield strength at temperature, defined as $R_y\sigma_{y,min}$ with R_y a trend line drawn through normalized, temperature dependent yield strength data. Typically, R_y is given by a fifth order polynomial fit as a function of temperature to the yield stress data normalized by the material yield strength at room temperature. S_u is the design tensile strength defined as $1.1R_u\sigma_{u,min}$ with R_u a trend line drawn through normalized, temperature dependent tensile strength data. R_u is defined similarly to R_y .

σ_r is the average creep rupture strength of the material. Though this is not an explicit Code rule, this strength is often defined in terms of a lot-centered Larson-Miller regression to rupture data [9]. S_r is the minimum strength to rupture, again typically defined as the 90% lower confidence interval of the same Larson-Miller correlation.

$\sigma_{minrate}$ is defined as the average stress to cause a minimum creep rate of 0.01% in $t/100$ hours, i.e. a minimum creep rate of 1% in t hours. For $t = 100,000$ hours the strict definition becomes the stress to cause a minimum creep rate of 0.01% in 1,000 hours, i.e. a minimum creep rate of 1% in 100,000 hours. Again, often this value is set with a lot-centered Larson-Miller correlation.

$\sigma_{1\%}$ is the average stress to cause 1% total strain in the given time. Again, this is often correlated and extrapolated from a lot-centered Larson-Miller model.

S_{tc} is the minimum stress to cause tertiary creep in the given time. Again, this is often correlated and extrapolated from a lot-centered Larson-Miller model and minimum is interpreted to mean a 90% lower confidence interval.

All four sets of allowable stress use broadly similar criteria, though the factors and the exact definitions vary. The following section analyzes each set of definitions, first considering the criteria on time-independent data, then time-dependent data, and then the overall definition of the allowable stress. Finally, the section concludes with a general analysis of the allowable stresses for set of design rules as a whole – accounting for the fact that Section III, Division 5 has two allowable stress checks, S_{mt} and S_o , both of which must pass for a component to pass the design rules.

For the time-independent regime, the main difference is in the tensile strength criteria. S_o and Section II, Part D, Table 1 (used by VIII-1) are the most conservative, followed by S_{mt} , followed by Section II, Part D, Table 5 (used by VIII-2, Class 2).

The time-dependent creep criteria include rupture and deformation data for all four allowable stresses, and the onset of tertiary creep only for S_{mt} . These values are extrapolated from shorter-term creep tests. Note there is a difference in the creep deformation allowable stress criteria with S_{mt} using time to a total strain of 1% and the others using the creep rate, often taken as the minimum creep rate. In general time to a total strain of 1% will be more restrictive as it includes the loading strain in a creep test (where the creep rate definition does not) and it includes the faster primary creep rate. In practice, these two criteria seldom control the allowable stresses in Section III, Division 5 and Section VIII.

Those differences aside, the Section VIII and S_o allowable stress definitions are less conservative on the rupture strength criteria compared to S_{mt} , with the Section II, Part D Table 1, Section II, Part D, Table 5, and S_o using 80% of the minimum rupture stress (S_r) while S_{mt} uses 67% of the minimum rupture stress (S_r). S_{mt} does not include the average rupture (σ_r) criteria included in the Section VIII and S_o allowable stresses.

S_{mt} at 100,000 hours will be lower than the corresponding Section VIII allowable stresses in the time-dependent regime.

Overall, the Section III, Division 5 allowable stresses will be no higher than the Section VIII allowable stresses for all conditions. For shorter design lives, the values of S_o will control design of a Section III, Division 5. S_o is nearly identical to the Section VIII, Division 1 allowable stress, meaning the Section III, Division 5 design will at minimum have the same margin as the Section VIII, Division 1 design. For longer design lives, S_{mt} will control the Section III, Division 5 design.

3.4 High temperature cyclic design

3.4.1 Summary of the design rules

The Section III, Division 5 Class A design rules provide a complete set of creep-fatigue and creep-ratcheting criteria. These aim to guard against creep-fatigue damage initiation and against explicit creep-ratcheting strain limits (1% average, 2% linearized bending, and 5% peak strain accumulation).

Section VIII, Division 1 does not address cyclic service and the design by rule approach does not consider cyclic failure modes in the design formula.

The Section VIII, Division 2, Part 4 design by rule formula do not consider fatigue or cyclic service. However, where cyclic service is expected, the design is required to demonstrate that the component will experience negligible fatigue per the Section VIII, Division 2, Part 5 rules. If the component cannot meet the screening criteria, then it must be analyzed to the Part 5 rules (or redesigned). If the component operates at high temperature, in the time-dependent regime of the allowable stress charts, then only the screening criteria based on comparable equipment is allowed, meaning this is the only method available to satisfy the screening criteria for high temperature components.

For fatigue, Section VIII, Division 2, Part 5 provides two general approaches: fatigue screening and a detailed fatigue analysis. Fatigue screening provides several different options for exempting the component from a detailed analysis:

1. Exemption based on experience with comparable equipment (5.5.2.2). “If successful experience over a sufficient time frame is obtained with comparable equipment subject to a similar loading histogram and addressed in the User’s Design Specification [...] then a fatigue analysis is not required as part of the vessel design.” The Code provides a detailed list of design features that should be considered when making this determination.
2. Method A (5.5.2.3), which counts pressure and temperature cycles, sums the cycles, and compares to general (i.e. not material specific) criteria. This approach cannot be applied to materials with room tensile strength exceeding 552 MPa.
3. Method B (5.5.2.4), which counts cycles but compares to limits based on material data.

Section VIII also specifies that in no case may a component that experiences more than one million load cycles be exempted from a detailed fatigue analysis.

For detailed analysis, Section VIII, Division 2 provides an elastic and elastic-plastic method for base metal and an elastic method for welds. These approaches are based on material-specific fatigue data provided in the Code.

For ratcheting Section VIII, Division 2 provides methods based on elastic and elastic-plastic analysis aimed at guarding against plastic ratcheting. These approaches do not consider creep-ratcheting or set specific strain limits.

3.4.2 Potential cyclic failure modes

In both Section VIII, Division 2, Part 5 and Section III, Division 5 the cyclic design rules consider two general failure modes: excessive strain accumulation and local damage initiation.

At low temperatures under repeated cyclic load, screening a component against plastic ratcheting is sufficient to limit the strain accumulated in a component. If the component does not ratchet, then it will accumulate finite strain over indefinite life. Both the Section VIII and Section III, Division 5 rules neglect any inelastic strain accumulated prior to shakedown or, more specifically, assume the component operates in the steady cyclic condition for the entire service period. This is reasonable as this offset strain is typically small [7].

However, at higher temperatures strain can accumulate in the component even if it operates in the classical shakedown regime [8]. This so-called creep-ratcheting strain might accumulate over time to cause gross distortion in the component geometry leading to actual failure or a service failure, where the distortion causes the component to not operate properly in the context of the plant systems.

Local damage initiation refers to the nucleation of a local fatigue or creep-fatigue crack. At low temperatures, this will typically occur via classical fatigue mechanisms [10] and can be designed against using typical engineering fatigue approaches [11].

At high temperatures, creep-fatigue or dwell fatigue can occur [12]. Both terms describe the observation that a standard strain-controlled fatigue test will often have a longer life compared to the same fatigue test with a hold at constant peak strain at the maximum compressive or the maximum tensile strain, or both. The exact mechanism causing this dwell effect is not clear and may vary from material to material [13, 14, 15, 16]. Numerous approaches have been proposed to guard against creep-fatigue failure over the past six decades [16, 17, 18, 19]. The ASME Section III, Division 5 rules adopt a D-diagram method where the designer calculates individual creep and fatigue damage fractions (i.e. assuming the component operates at quasi-steady conditions and pure fatigue conditions, respectively) and the consults an interaction diagram, determined from experimental creep-fatigue test data [20], to decide if the component is acceptable.

3.4.3 Limitations of the Section VIII rules

The Section III, Division 5 rules are intended to account for both elevated temperature cyclic failure mechanisms – creep-ratcheting and creep-fatigue.

None of the Section VIII rule sets considered here fully account for high temperature cyclic failure.

The Section VIII, Division 1 rules do not account for fatigue, even at low temperatures.

The Section VIII, Division 2, Part 4 rules require the designer to apply the “comparable service experience” screening criteria. Potentially international experience with high temperature reactors, for example the advanced gas reactor fleet in the United Kingdom, might provide a basis to apply this screening criteria [21].

This leaves Section VIII, Division 2, Part 5. We cannot evaluate the basis for the Method A screening test. Essentially, this provides a simple limit on the number of cycles experienced by a component. Without a detailed evaluation of the basis behind these cycle counts the authors cannot determine if this test might reasonably apply to high temperature nuclear components.

This leaves a full fatigue analysis and the Method B screening test. These approaches all rely on the fatigue data provided in Annex 3-F. This fatigue data does not extend into the creep regime – the Annex explicitly prohibits the use of the data above 371 °C or 427 °C, depending on the material, which is the classical ASME threshold for elevated temperature service.

As such, this report concludes that the current Section VIII rules offer one path, “comparable service experience”, for evaluating fatigue at high temperatures. Chapter 5 briefly discussed potential mitigation, which include using the existing Section III, Division 5 high temperature fatigue data – an approach adopted in the next chapter – or requiring designers or owners to develop their own high temperature fatigue data.

Additionally, the design by analysis approaches for fatigue and ratcheting provided in Section VIII, Division 2 do not account for the specific high temperature cyclic mechanisms described above: creep-ratcheting and creep-fatigue. Instead, the methods are geared towards conventional, low temperature design where evaluating plastic ratcheting and classical fatigue are sufficient.

Again, the final chapter of this report lists potential mitigation approaches. One approach might be to expand the current Section VIII design rules to account for these mechanisms. An effort is underway at ASME to consider such design approaches for both Section I and Section VIII.

An alternative would be to develop more suitable negligible fatigue criteria, accounting for high temperature material properties and failure modes, and to develop similar negligible creep criteria. If a component operates under negligible fatigue, then the existing Section VIII protections against steady creep rupture via the allowable stresses might be sufficient to provide an adequate design. Similarly, if the component operates under negligible creep criteria, then the existing Section VIII, Division 2 fatigue design rules could provide an adequate design.

Table 3.3 summarizes the key limitations identified for the Section VIII cyclic design rules, if applied to high temperature nuclear components.

Table 3.3. Summary of limitations identified in the Section VIII cyclic design rules.

Limitation	Applicability
Lack of fatigue criteria, even at low temperatures	VIII-1
Requires component to meet fatigue screening criteria for elevated temperature service	VIII-2 Part 4
Fatigue screening criteria not applicable/not well supported at elevated temperatures	VIII-2 Part 4 and Part 5
Detailed fatigue evaluation does not account for creep-fatigue interaction	VIII-2 Part 5
Joint efficiency factors do not account for the experimentally-observed decrease in weld cyclic strength, even for defect-free weldments	VIII-2 Part 4 and Part 5

3.5 Allowable materials

The material library available to designers working under the four different sets of Code rules is one immediately noticeable difference. The Section III, Division 5, Class A rules only permit six materials: 304H and 316H stainless steel; austenitic 800H; Grade 91 ferritic-martensitic steel; low alloy Grade 22 steel, in the annealed condition; and the nickel-based Alloy 617 (additional materials are available for Class B construction, though not for welded construction operating in the elevated temperature regime for long periods of time). By contrast, Section VIII provides a large catalog of materials including plain steels, low alloy steels, austenitic and duplex stainless steels, nickel-based alloys, and many other materials. With a few exceptions, the Section III, Division 5³ materials are all available for Section VIII design.

However, as noted above the larger Section VIII material library is constrained by the lack of high temperature cyclic data. This data does not exist in the ASME Code, except for the six Class A materials. However, gathering high temperature fatigue data may be the only mitigation strategy when applying the Section VIII design rules to a component that will experience elevated temperature cyclic service. Collecting the additional data often used to parameterize high temperature design methods like creep-fatigue testing and full creep curves may also be necessary.

3.6 Treatment of welds

Both the Section III, Division 5 and Section VIII rules allow for welded construction. The rules distinguish between product forms that may include welds (i.e. welded pipe and tubing) from welded construction, i.e. fabricating components and making connections with welds in the fabrication shop or in the field. A simple way to think of the difference between the two types of welds is that welded product forms will come to the fabricator with the welds already there, whereas fabrication welds will be made to join parts made from base material products.

Both sets of rules can account for differences in the strength of welded product forms through the tabulated allowable stresses but account for the effects of fabrication welds via the design rules, often by reducing the strength of a (fabrication) weldment from the allowable stress of the underlying base product form. The subsequent discussion focuses on fabrication welds, not welded product forms, where the different design rules under consideration adopt very different techniques for accounting for the strength of (fabrication) weldments.

³ For reasons that are not clear to the authors, the allowable stresses in Section II, Part D, Table 5 for Section VIII, Division 2 design do not include plate 316H stainless steel as an allowable product. This table does, however, include 316H stainless steel forgings. Moreover, the 316H stainless steel allowable stresses in Table 5 seem anomalously low in the high temperature regime, for example lower than the values provided for 304H stainless steel. This difference is in the time-dependent regime and therefore cannot be explained by the different minimum tensile strength products available from the ASME SA-240 and SA-182 specifications. Alloy 617 is not available for Section VIII, Division 2 high temperature design, which is less surprising as it is not a commonly used material in non-nuclear construction. Finally, Grade 91, Type 2 is not currently available in Table 5A (only Type 1 is available). Section III, Division 5 uses Type 2 as the material for Class A construction.

The design by rule Section VIII, Division 1 and Section VIII, Division 2, Part 4 methods account for the effects of welds through the specific design formulas, which often specify the types and geometry of permissible welds. Evaluating the breadth of these design by rule formula is beyond the scope of this report.

Section VIII does not alter the allowable stresses for welds or welded construction. This tacitly assumes an overmatched weld in the time-independent and time-dependent regimes. The Section VIII rules do apply a “joint efficiency factor” E to the allowable stress for welded construction, which may be less than 1. However, this joint efficiency factor is not based on test data, nor does it account for differences in the material strength of weldments versus base metal. Instead, this efficiency factor varies by type of weld, the type of weld material, and by the non-destructive examination technique applied to the completed weld. For most types of welds a full RT or UT inspection provides a factor of 1.0 (i.e., no reduction in the allowable stress). The joint efficiency factor therefore accounts for the possibility of a defect in the final weldment, rather than any reduction in strength of a properly-formed weld.

Section VIII, Division 2, Part 5 does not reference these joint efficiency factors. Historically, Section VIII, Division 2 required full radiographic inspection of welds and hence a joint efficiency of 1.0 [22]. However, more recent editions of Section VIII, Division 2, Part 4 (design by rule) call out the joint efficiency factor and allow for values less than 1.0. Similar changes to Section VIII, Division 2, Part 5 have not been completed, but might be relatively easily made [22]. The Code does not explicitly specify, but a reasonable interpretation is that Part 5 can only be applied to joints that have an efficiency factor of 1.0, i.e. the most rigorous inspection criteria per Table 7.2.

By contrast the Section III, Division 5, Class A design rules provide stress rupture factors to reduce the allowable stresses for welds in the time-dependent regime. The Section III, Division 5 rules assume overmatched welds in the time-independent regime but admit the possibility of undermatched welds in the creep regime. Moreover, Class A welded construction generally requires a double volumetric inspection of all welds, greatly reducing the possibility of a defect in the weldment under service conditions. The stress rupture factors, therefore, account for the difference in strength of the weldment, rather than the potential presence of a defect while the inspection rules screen out defected weldments.

Data suggests that weldments operating in the creep regime often have significantly shorter creep rupture lives compared to the component base materials. Section I also provides for the reduced strength of weldments at high temperatures through the use of Weld Strength Reduction Factors (WSRF) conceptually similar to the Section III, Division 5 stress rupture factors. However, Section VIII does not account for the reduced creep strength of weldments. This may contribute to a reduced design margin for Section VIII components operating in the creep regime, particularly where welds are located in high temperature, high stress areas of the component.

Section VIII, Division 2 does consider the potential for reduced fatigue strength of weldments, compared to base material, by providing a specific evaluation procedure for fatigue in welded connections. However, this method is not applicable at high temperatures as the Code does not provide design data for high temperatures.

In addition to the use of the stress rupture factors in the creep damage calculation, Section III, Division 5 also reduces the pure fatigue strength of welds by a factor of one-half, compared to base material. While this approach is crude compared to the Section VIII evaluation approach, it conservatively accounts for the reduced fatigue strength of weldments without needing additional, weld-specific, fatigue data.

Section III, Division 5 also reduces the allowable strain accumulated in weldments by a factor of one-half, compared to base material. The Section VIII strain accumulation and ratcheting rules do not distinguish weldments from base metal, beyond the use of the efficiency factor.

Similar to the availability of base materials detailed above, Section VIII does provide a much wider assortment of allowable weld types (filler metal/weld procedure combinations). Only a few weldments are available for Class A construction, likely limited by the need to collect weld metal and/or weldment creep data to develop stress rupture factors. By contrast, Section VIII makes available a large number of weld types, with the main restriction simply being the need to qualify the weld per the Section IX rules.

3.7 Types of components

The Section III, Division 5, Class A rules cover essentially all types of components found in high temperature reactors: vessels, nozzles, heat exchangers, pumps, valves, and piping. By contrast, Section VIII only covers “vessels.” However, the Section VIII rules interpret “vessels” liberally, covering nozzles, heat exchangers, and several other component types. However, the Section VIII rules do not cover piping.

For high temperature piping systems, designers would need to turn to another design code, likely B31.1 or B31.3. A separate evaluation of the piping rules versus Section III, Division 5 would be needed to determine differences in the design rules and design space, analogous to this report.

4 Evaluation of design space and comparative margin with the Bree cylinder

4.1 The design problem

This chapter provides a quantitative comparison of the feasible design space and comparative margin of the Section III, Division 5; Section VIII, Division 1; Section VIII, Division 2, Class 2, Part 4; and Section VIII, Division 2, Class 2, Part 5 rules.

By feasible design space this report means the region of design space a component passes all the relevant design rules. By design space this report means a parametric description of both the component geometry and the loading applied to the component. For example, a particular component might be thicker or thinner, it might have more or less pressure or thermal stress, it might have a longer or shorter hold time at steady conditions, etc.

The Bree cylinder [7] provides a simple, yet fairly comprehensive, component to consider for this design study. The Bree cylinder is a fictitious configuration, as it is typically represented as an open-ended cylindrical pressure vessel under a combination of a constant pressure and an alternating, through-wall thermal gradient (Figure 3.1). For the standard Bree cylinder, the pressure is constant and the temperature alternates from a constant profile across the wall thickness at the start of the load cycle, to a linear profile during a hold at fixed conditions, back to a constant profile to repeat the cycle. During this temperature cycle, the inner wall temperature remains constant and the outer wall temperature varies.

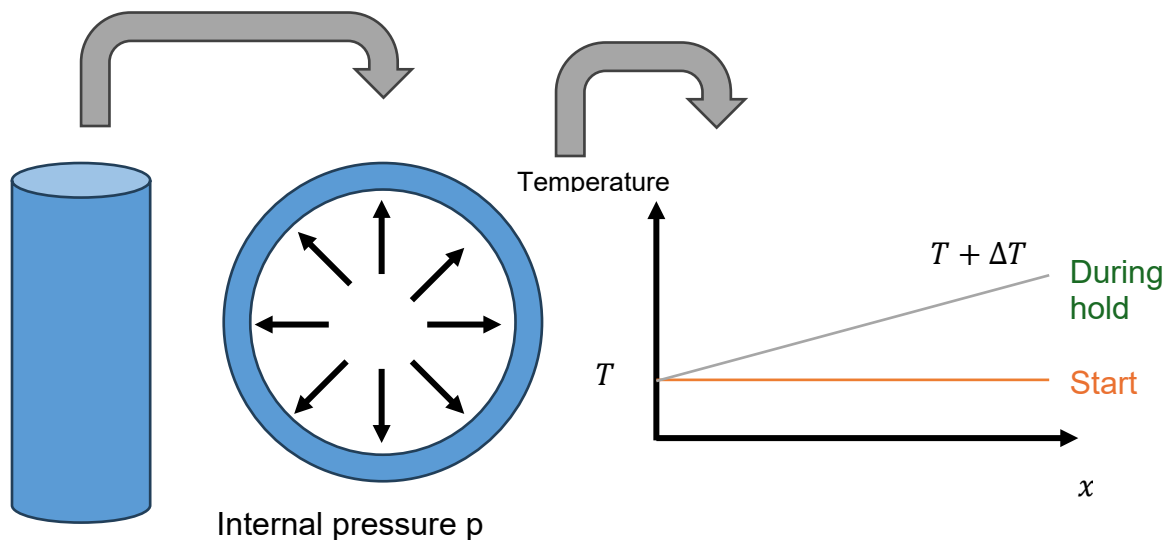


Figure 3.1. The standard interpretation of the Bree problem as an open-ended cylindrical vessel. In this diagram T is the design temperature, ΔT is the thermal gradient, and x is the through-wall coordinate from the vessel outer to inner radius.

An open-ended cylinder under internal pressure is not a realistic component configuration. Despite this, the Bree problem forms the basis of much of the Section III and Section VIII strain accumulation and ratcheting rules. There are at least two reasons for its widespread use:

1. The original paper defining the geometry and loading argues that not considering the axial stress is bounding, i.e. the open-ended vessel will ratchet more quickly compared to a closed vessel.
2. Solutions to the general Bree problem are often generalized to arbitrary components under arbitrary loadings via the Bree diagram.

Figure 4.2 shows the classical Bree diagram included in the ASME Code for the constant-pressure, alternating temperature problem. Bree diagrams have a non-dimensionalized primary load (P) plotted on the x-axis and a non-dimensionalized secondary load (Q) plotted on the y-axis. The labeled regions on the chart show different types of cyclic behavior experienced by the component for different combinations of primary and secondary load: pure elastic, elastic shakedown, plastic shakedown, and ratcheting. The number, type, and shape of the regions will vary based on the specific problem under consideration. However, this diagram for the classical problem defined above is widely used in ASME and other design methods to represent arbitrary components, not just the specific Bree cylinder. Design methods generalizing the diagram enter the x- and y-axes with stresses given by classifying elastically-calculated stresses into primary and secondary categories and then converting the stress to a uniaxial measure, for ASME using the stress intensity (maximum shear). Later work extended the concept to account for creep-ratcheting [8] and the diagram (or more specifically the core stress derived from a Bree analysis) forms a key basis for the Section III, Division 5 design by elastic analysis rules for creep-fatigue.

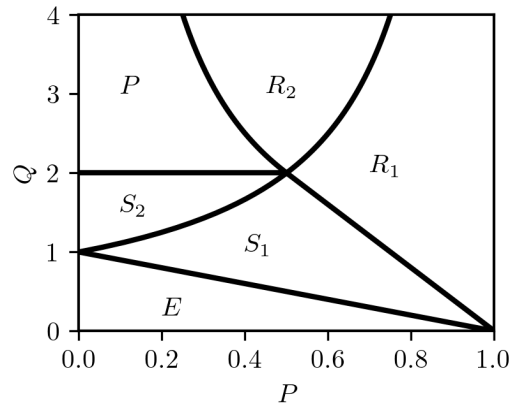


Figure 4.2. The classical Bree diagram. Here the labels and lines indicate regions of pure elastic behavior (E), elastic shakedown (S), plastic shakedown (P), and ratcheting (R). The subscripts indicate different analytic solutions for the resulting steady cyclic stresses, though the basic shakedown behavior in each region sharing the same letter is the same.

We use the Bree cylinder in this report to provide a concrete design to analyze with the four sets of design rules because it generalizes to cover a wider variety of design situations. Moreover, because the cylinder is a comparatively simple geometry, the authors can easily parameterize the design space and run many design calculations quickly.

The key parameters for the Bree problem are:

1. The normalized primary stress $P = pr/(tS_y)$ with p the applied pressure, r the vessel radius, t the vessel thickness, and S_y the (design) yield strength.

2. The normalized secondary stress $Q = \alpha E \Delta T / S_v$ with α the coefficient of thermal expansion, E the material Young's modulus, and ΔT the linearized, through-wall, thermal gradient.
3. The operating temperature, T .
4. The design life of the component, t_{design} .
5. The length of the hold at the steady state, t_{hold} . The standard Bree problem keeps the pressure fixed and alternates the thermal gradient between the linear through-wall gradient and an isothermal condition. This provides a driving force for creep-fatigue damage. Alternative problems include alternating pressure, but the classical problem with constant pressure is often used because assuming constant pressure simplifies the stress analysis.

The number of repetitions of the alternating load (here the temperature gradient) over the design life is another key parameter, but it can be calculated from these five values. For simplicity, this analysis ignores the time required to ramp the temperature gradient from zero/isothermal to the full value and so the number of repetitions is $N = t_{design}/t_{hold}$. Also, for simplicity the analysis uses material properties and design data at a constant temperature, T , rather than vary these properties with the application of the alternating thermal stress. This is a reasonable approximation, as allowable steady state temperature gradients are typically small, meaning the temperatures will not vary significantly from the constant temperature.

We can complete a comparative parameterized Bree analysis for four materials: Grade 22, 316H, 304H, and Alloy 800H. As noted above, design allowable stresses for Section VIII, Division 2 do not exist for Alloy 617 and the allowable stresses for Grade 91 for Section VIII, Division 2 are for Type 1 material, not the Type 2 used in Section III, Division 5. This report includes results for 316H and Grade 22 as the results for these materials are representative of the other two materials. This analysis uses the plate allowable stresses for all materials, except for 316H for Section VIII, Division 2 the analysis uses the allowable stresses for forgings as the plate product form is not included in the allowable stress table.

For each material, this analysis selects a range of design parameters. Table 4.1 lists these parameters for the two materials in this report. The study considers the entire Cartesian grid of all combinations of the discrete parameters listed in this table, meaning each study for each material encompasses 400,000 independent designs. For 316H the 2023 edition of the Code imposes a maximum 300,000-hour design life for 316H. We used draft 500,000-hour allowable stresses under consideration by ASME to explore the differences in the design rules for very long design lives.

Table 4.1. Design conditions for the Bree analysis.

	P	Q	T	t_{design}	t_{hold}
Grade 22	0 to 1 in 10 equal steps	0 to 4 in 20 equal steps	400 to 600 °C in 20 equal steps	10,000 to 300,000 hours in 10 equal steps on a log scale	10 to 10,000 hours in 10 equal steps on a log scale
316H	0 to 1 in 10 equal steps	0 to 4 in 20 equal steps	450 to 800 °C in 20 equal steps	10,000 to 500,000 hours in 10 equal steps on a log scale	10 to 10,000 hours in 10 equal steps on a log scale

To generate a point in this design space, the Bree cylinder is analyzed for the given material and given combination of parameters. The details of these design checks for each set of Code rules considered are in the next section. The result of this calculation is a pass or fail answer – either the design for this material and set of parameters passes all the relevant Code rules or it fails. We then summarize the results of these 400,000 analyses in the final section of this chapter.

4.2 Design calculations

This section summarizes the design calculations followed for each of the four sets of rules to give a pass/fail answer for a particular Bree cylinder design. For the design by analysis options (Section III, Division 5 and Section VIII, Division 2, Part 5) the design by elastic analysis options are used – both sets of rules provide options for elastic, perfectly plastic analysis and Section III, Division 5 further provides full inelastic analysis methods. No stress analysis is required to calculate the elastic stresses for the Bree cylinder, instead, the non-dimensionalized stresses can be multiplied by the material yield strength to give the actual value of stress for a given condition.

For the design by rule approaches (Section VIII, Division 1, and Section VIII, Division 2, Part 4), the design by rule formula for a cylindrical shell was slightly altered, as outlined below. However, for both these sets of rules, for thin shells with $\frac{r}{t} > 10$, the approximations introduce minimal differences between the original design formula and the version used here.

This example is for non-welded construction, so no stress rupture factor is applied for the Section III, Division 5 rules and an efficiency factor of $E = 1$ is used for the Section VIII methods. The equations below include the joint efficiency factor, even though it does not apply here, because that is how the allowable stress equations are presented in the Code.

4.2.1 Section VIII, Division 1

UG-27 gives the design by rule formula for a cylindrical shell as

$$p = \frac{SEt}{r + 0.6t}$$

with S the allowable stress, E an efficiency factor taken to be $E = 1.0$ for these calculations, r the vessel radius, t the vessel thickness, and p the pressure. This formula is the so-called “ASME approximation” to the Lamé solution for thick-walled vessels. In the limit of thin-shell theory this formula reduces to

$$p = \frac{SEt}{r}$$

i.e., the classical thin-shell formula. We can cast this as a design check simply by noting

$$PS_y = \frac{pr}{t}$$

and so, the design check is

$$PS_y \leq SE$$

No additional checks are required for the simple Bree geometry. This allowable stress check does not impose a limit on the secondary stress nor on the number of load cycles.

4.2.2 Section VIII, Division 2, Part 4

Section 4.3.3 provides the basic design by rule formula for a cylindrical shell as

$$t = r \left(\exp \left[\frac{p}{SE} \right] - 1 \right)$$

which is the Lamé solution for a thick-walled vessel. For thin wall theory this equation reduces to the same allowable stress check as for Section VIII, Division 1

$$PS_y \leq SE$$

albeit with a different allowable stress.

The Code additionally requires that no section be less than 1.6 mm thick. We did not consider this criterion here.

For design by rule, for Class 2 construction, the designer is required to complete the fatigue screening test in Part 5. For components operating in the creep regime, the only acceptable test is the screening analysis based on experience with comparable equipment (4.1.1.4). As this analysis cannot apply such a criterion to the Bree problem, the authors (incorrectly, per the Code) applied the 1,000-cycle limit contained in Section 5.5.2.3 to provide some sort of fatigue screening, even though the Code specifically disallows this fatigue screening test at high temperatures.

4.2.3 Section VIII, Division 2, Part 5

Our analysis considers three criteria from Section VIII, Division 2, Part 5.

4.2.3.1 Primary load limit

For the Bree problem the primary load limit is simply

$$PS_y < S$$

for the appropriate allowable stress S .

4.2.3.2 Ratcheting limit

Our analysis considers the elastic limit given in Section 5.5.6. For the Bree cylinder, this check reduces to two conditions. If the temperature falls into the italicized values of the allowable stress tables in Section II, Part D, i.e. the allowable stresses controlled by time-dependent properties, then the check is

$$PS_y + QS_y < 3S$$

Otherwise, the check is

$$PS_y + QS_y < \max\{3S, 2S_y\}$$

4.2.3.3 Fatigue limit

This analysis implements the fatigue check in Section 5.5.3.2. To execute this check, high temperature fatigue data is needed, but it is not available for any material in Section VIII. To provide this data, the design fatigue curves in Section III, Division 5 are transformed. The Section VIII elastic fatigue methods work with the alternating stress S_{alt} , whereas the Division 5 fatigue data works with the strain range. To convert, an elastic strain rate is calculated from the alternating stress intensity with the formula

$$\Delta\varepsilon = 2S_{alt}/E$$

This strain is used to index into the Division 5 design fatigue curves to give an allowable number of cycles, N .

The design check in Section 5.5.3.2 reduces to, for the Bree cylinder:

$$N \leq N_{allow}(S_{alt}, T)$$

with N_{allow} the allowable number of cycles for a given alternating stress and temperature, using the approach above, and

$$S_{alt} = K_v QS_y / 2$$

with

$$K_v = \frac{1 - \nu}{1 - \nu_p}$$

where ν is the Poisson's ratio for the material from Section II, Part D and

$$\nu_p = \max\left\{\nu, 0.5 - \frac{S_y}{S_{alt,allow}}\right\}$$

where $S_{alt,allow}$ is the maximum allowable alternating stress for temperature T and N load cycles, i.e. the alternating stress that results in an allowable number of cycles N for temperature T .

4.2.4 Section III, Division 5, Subsection HB, Subpart B

We completed the Section III, Division 5, Class A analysis with the hbbanalysis software package [23] developed by Argonne National Laboratory for the NRC. This section does not rehash the equations giving the actual design limits, as those are well described in the documentation for the hbbanalysis package. This section briefly outlines the path taken for this analysis through the various alternative approaches in Section III, Division 5.

Again, the overall design check considers three design limits: primary load, ratcheting strain accumulation, and creep-fatigue damage.

1. The primary load limits are defined in terms of the allowable stresses S_o and S_{mt} . The criterion for a Bree cylinder is the same as for Section VIII, Division 2, Part 5 above, i.e.

$$PS_y \leq S$$

except $S = S_o$ for the design check and $S = S_{mt}$ for the service check. As S_{mt} invokes the component design life the primary load limits now include the time expected in service.

2. We applied test B-1 defined in HBB-T-1332. This is the O'Donnell-Porowski approach, using a Bree analysis supplemented by a creep analysis with isochronous stress-strain curves to account for creep-ratcheting.
3. We apply the HBB-T-1432(e) criterion for calculating the plastic/creep corrected strain range. However, a resetting relaxation history is assumed, not taking advantage of the option in HBB-T-1433(b) even if the design meets the criteria for global relaxation.

4.3 Results and discussion

It is difficult to summarize the results of 400,000 designs covering the five key variables in a limited number of plots. We elect to display the results of the calculations on Bree diagrams where the x-axis is the normalized primary load and the y-axis is the normalized secondary load. The normalization for each dimension is described in the previous section detailing the Bree diagram. These diagrams provide an envelope of acceptable designs – literally the feasible design space – for a fixed temperature, design life, and hold time. These plots clearly display the feasible space for fixed conditions.

To account for the other three variables (temperature, design life, and hold time), several series of these Bree diagrams are plotted:

1. Series 1: Fixed design life of around 100,000 hours, fixed hold time of 10,000 hours, increasing temperature. This shows the effect of temperature on the feasible space for all the rules as well as how varying temperature changes the relative margin between the different design options. We pick a design life of 100,000 hours for the best comparison with the Section VIII rules, where allowable stresses are based on extrapolated 100,000 hour properties. We pick a hold time of 10,000 hours to represent a traditional refueling cycle of around one year.
2. Series 2: Fixed hold time of 10,000 hours, pick two fixed temperatures for each material (a “cooler” and a “hotter” temperature) and vary the design life. This series shows how the design life affects the feasible space.
3. Series 3: Fixed design life of around 100,000 hours, pick two fixed temperatures for each material (again cooler and hotter) and vary the hold time. This series examines how the design rules respond to plants that experience more transients with short durations (for example, load following systems adjusting the plant output to react to grid conditions) versus more traditional operation with fewer transients with longer durations.

Numerous additional comparisons are possible and these views into the results show only a narrow slice of the 400,000 results. Additional visualization and processing might reveal additional trends, beyond those summarized below.

4.3.1 Series 1: Effect of temperature

Figure 4.3 plots the temperature series for 316H and Figure 4.4 for Grade 22. These plots keep the design life and the hold time fixed and vary the temperature sequentially.

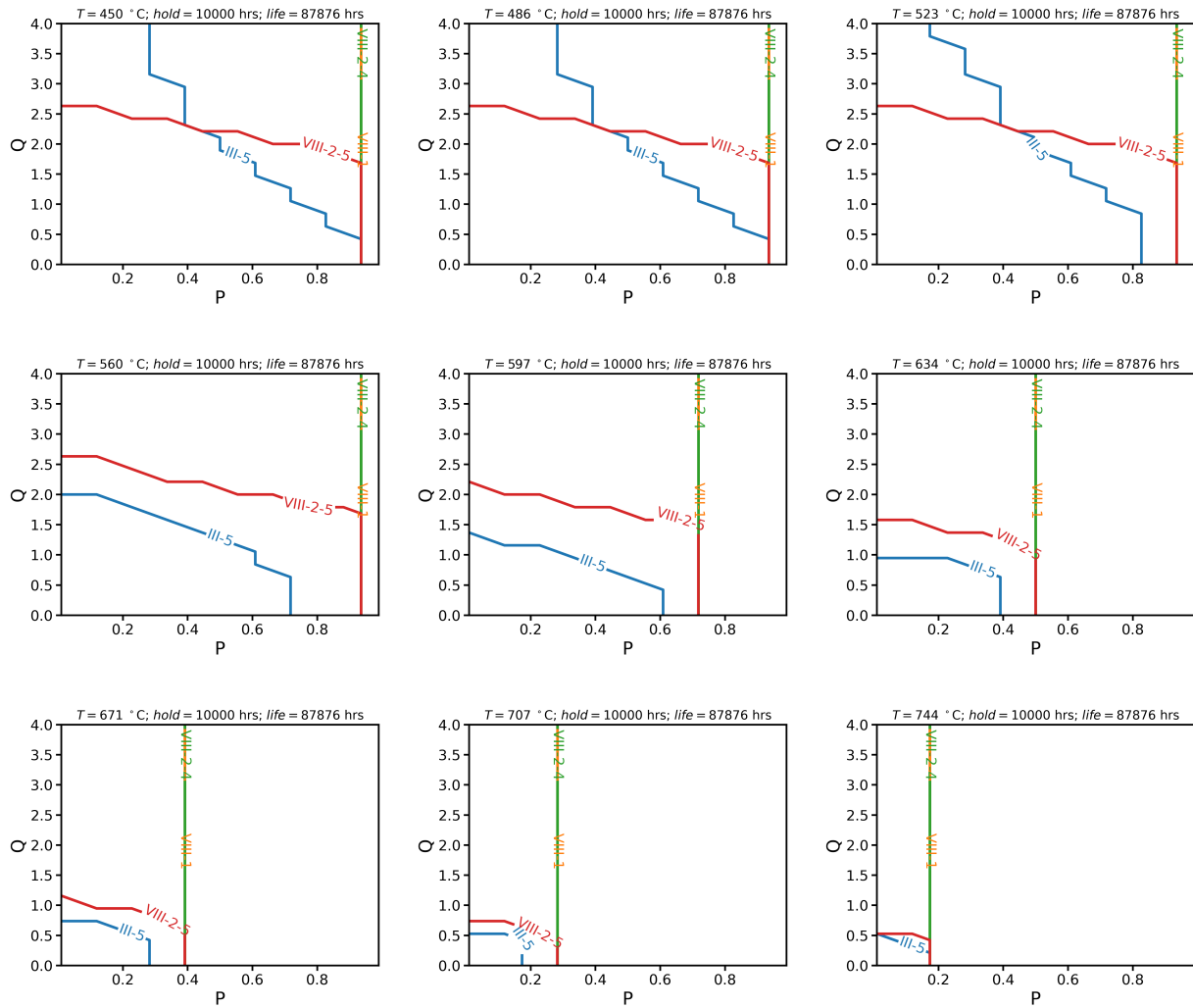


Figure 4.3. Temperature series of design space charts for 316H.

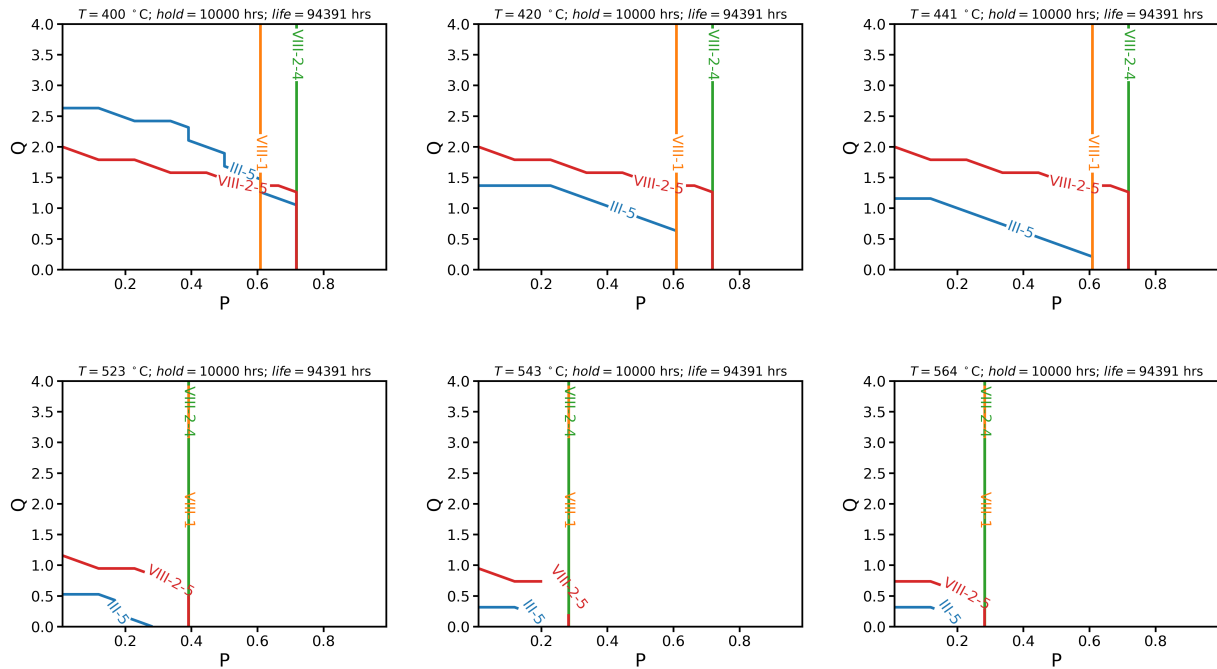


Figure 4.4. Temperature series of design space charts for Grade 22.

As expected, the Section VIII-1 and Section VIII-2, Part 4 rules provide a design envelope uncapped on the secondary load axis. This is because these sets of rules have no limit on the secondary stress or the number of repetitions of the load cycle.

Along the primary load axis, the design by rule and design by analysis options all provide a similar design limit, except for Grade 22 in the lower temperature regime for Section VIII, Division 1. The allowable stress for the Section VIII, Division 1 rules and the S_o allowable stress for Section III, Division 5 is lower than the allowable stress for Section VIII, Division 2 and S_{mt} in the time-independent regime. However, these differences disappear at higher temperatures. The main difference between the approaches is that the design by analysis methods provides a cap on the allowable alternating secondary (thermal) stress.

The only difference between the Section VIII-1 and Section VIII-2, Part 4 is the definition of the allowable stress. At high temperatures in the creep regime both methods use same criteria to define the allowable stress and therefore produce the same design envelope. At lower temperatures the Section VIII-2, Class 2 allowable stresses provide a higher allowable stress for materials controlled by the tensile, as opposed to yield, strength. Here, Grade 22 falls into this category but 316H does not.

In general, the Section III, Division 5 rules provide a more restrictive feasible design space. However, at lower temperatures the secondary stress limits imposed by the collective Section III, Division 5 rules allow for less restrictive designs compared to Section VIII, Division 2, Part 5. This is because at low temperatures the material accumulates comparatively little creep-fatigue damage, meaning the ratcheting strain accumulation criteria control the design. The Section VIII,

Division 2, Part 5 rules are approximately based on limiting the secondary stress to twice the yield stress; a criterion aimed at preventing plastic shakedown (see Figure 4.2). The cap for this method along the normalized secondary load axis is about 2.0 – slightly greater because the actual design limit references the allowable stress and not the yield stress itself. By contrast, the Section III, Division 5 rules allow components to operate in the plastic shakedown regime at alternating stresses greater than twice yield, provided the component does not accumulate excessive creep strain and/or creep-fatigue damage. This means the Section III, Division 5 design space can extend up to the plastic shakedown limit (again, see Figure 3.2) in the case where the design accumulates very little strain per cycle and avoids reaching the strain accumulation limit.

4.3.2 Series 2: Effect of design life

Figure 4.5 and Figure 4.6 plot the design life series for 316H for a lower and higher temperature, respectively. Figure 4.7 and Figure 4.8 are the same plots for Grade 22.

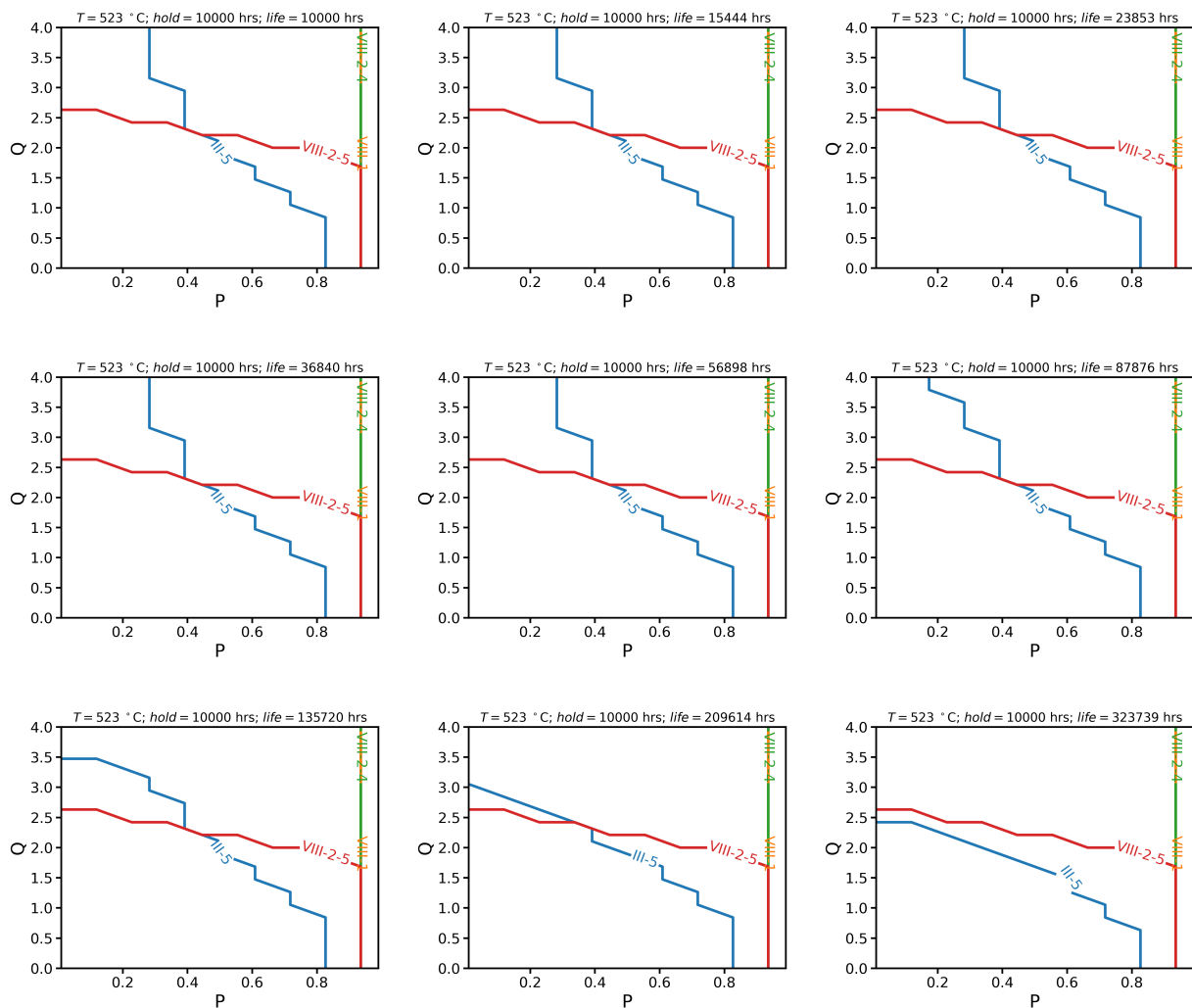


Figure 4.5. Design life series for 316H at a cooler temperature.

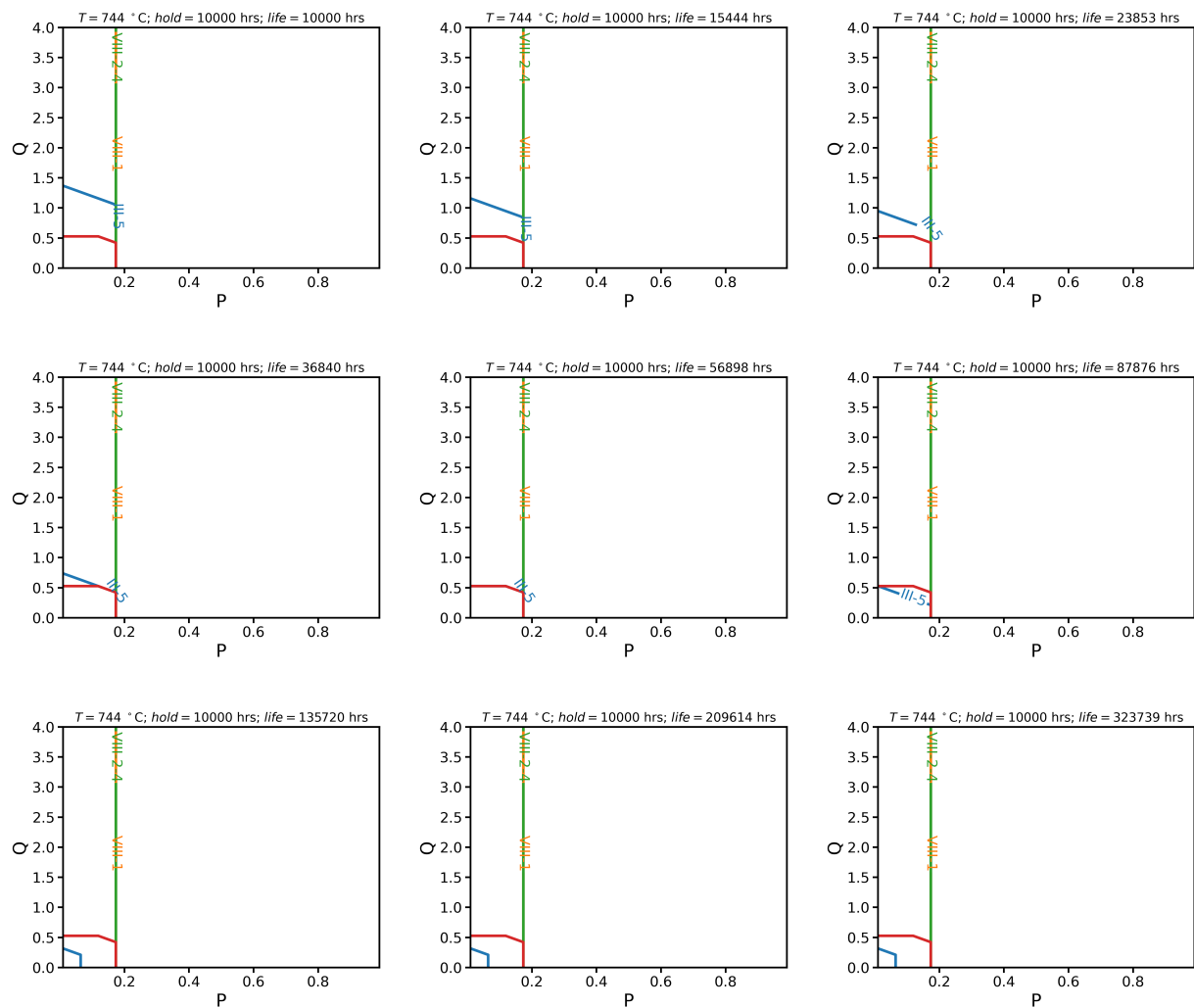


Figure 4.6. Design life series for 316H at a warmer temperature.

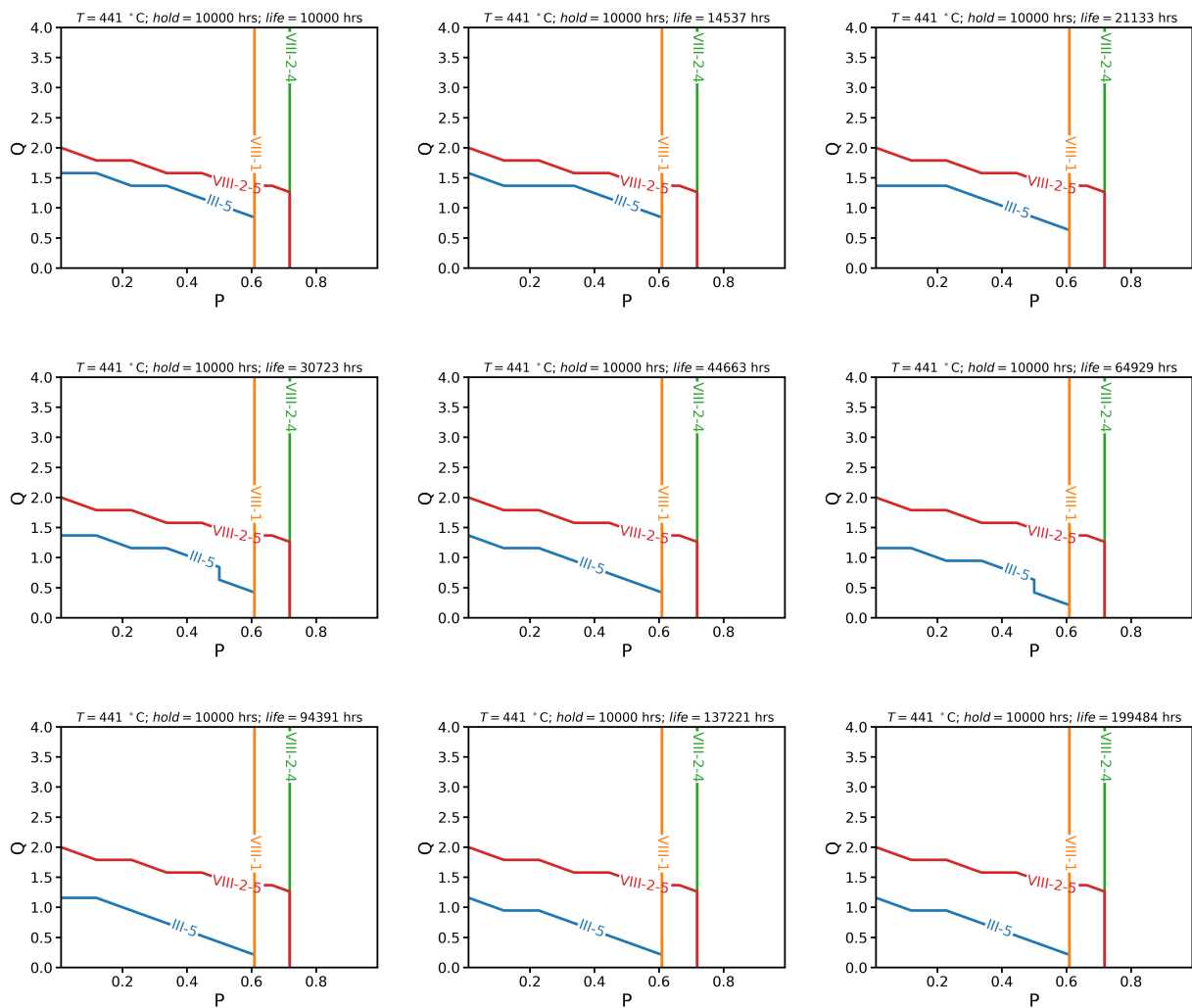


Figure 4.7. Design life series for Grade 22 at a cooler temperature.

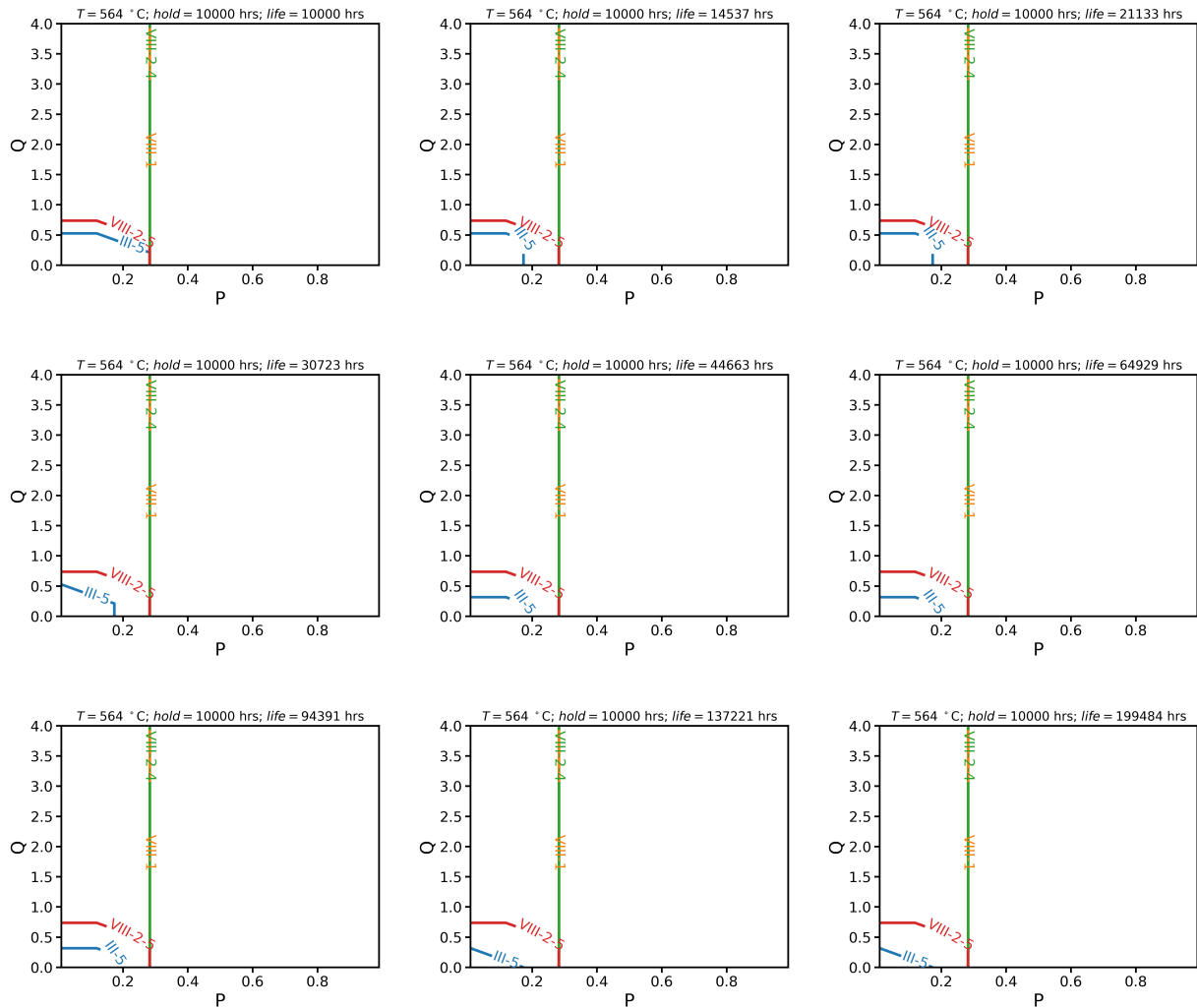


Figure 4.8. Design life series for Grade 22 at a warmer temperature.

The Section VIII design methods are nearly entirely insensitive to design life. This is expected for primary load, as the primary load allowable stress is independent of any design life. We might expect the feasible design space to decrease on the secondary load axis as design life increases only because the number of cycles experienced by the component is increasing. We do not see this trend for this problem because the ratcheting criteria control the secondary stress limit. Since the Section VIII, Division 2, Part 5 design rules only consider pure fatigue a component must experience many fatigue cycles to accumulate significant damage.

The remainder of the observations mirror the temperature series. We can add “short life” to “low temperature” to list of conditions when the Section III, Division 5 rules are less restrictive than the Section VIII, Division 2, Part 5 rules along the secondary load axis. However, for higher temperatures or longer design lives the Section III, Division 5 rules rapidly become more restrictive. Again, the same pairings along the primary load axis are observed as noted in the previous section, related to the definition of the allowable stress.

4.3.3 Series 3: Effect of hold time

Finally, Figure 4.9 and Figure 4.10 and Figure 4.11 and Figure 4.12 show the hold time series for 316H, at low and high temperature, and for Grade 22, again at low and high temperature.

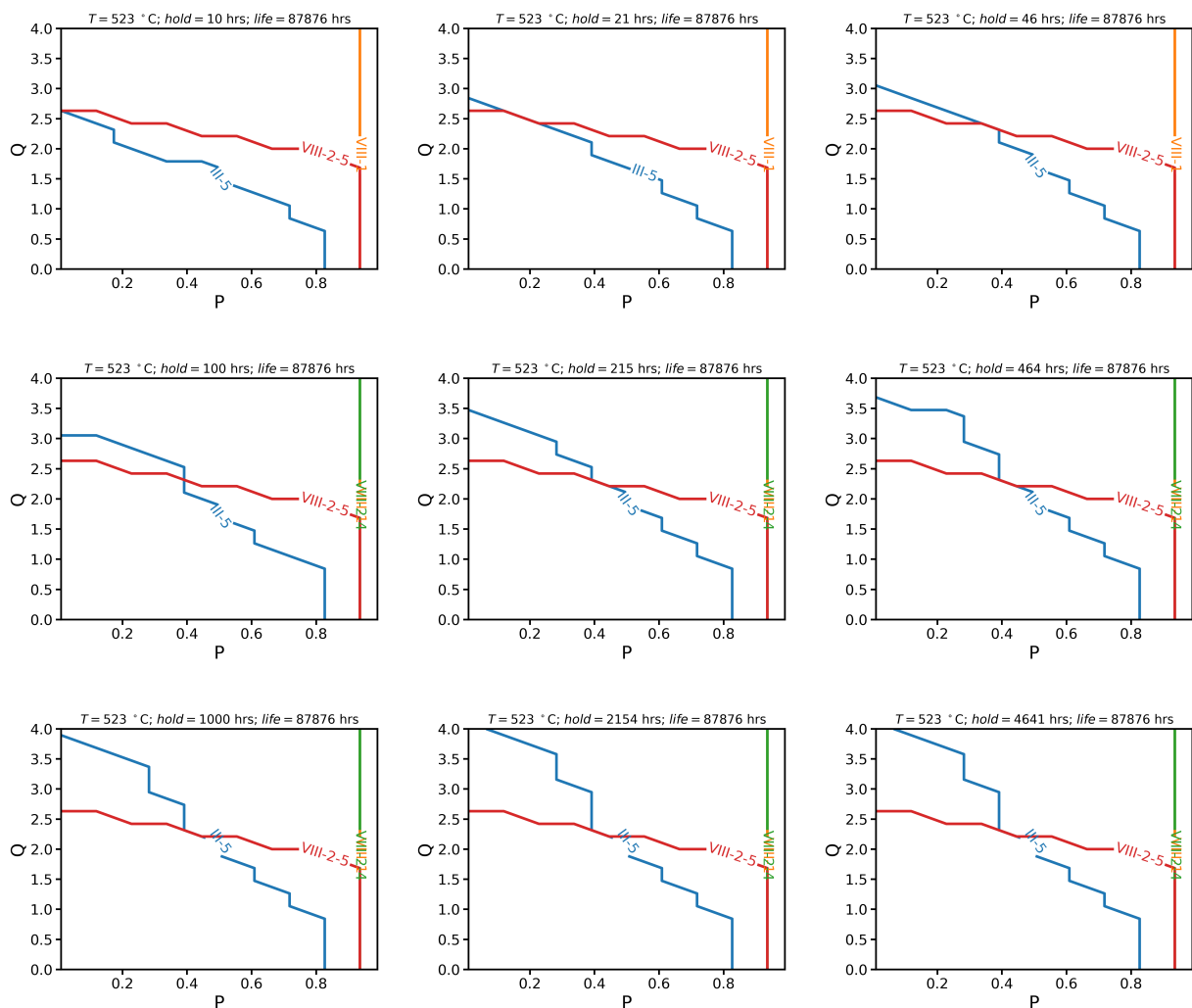


Figure 4.9. Hold time series for 316H at a cooler temperature.

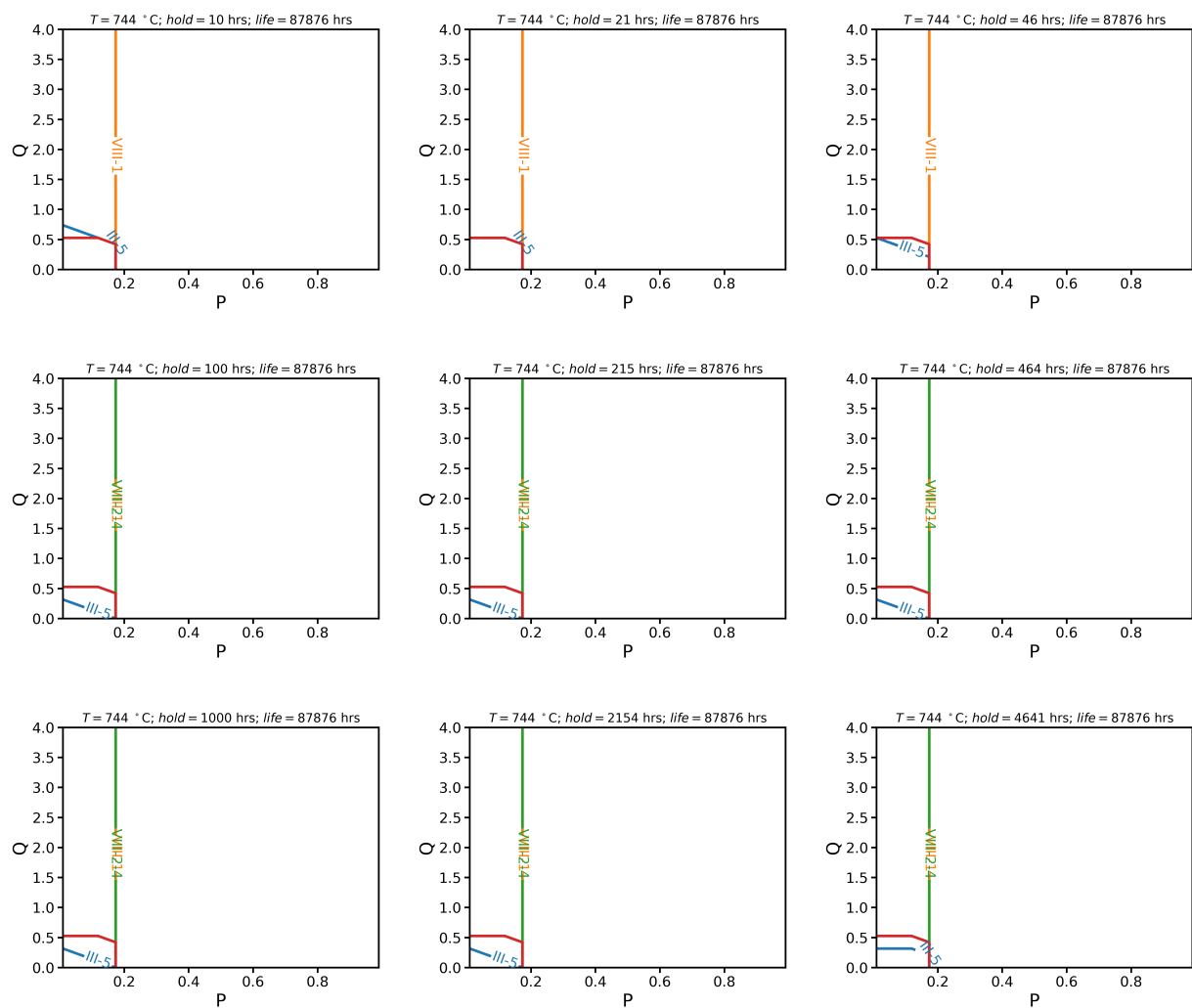


Figure 4.10. Hold time series for 316H at a warmer temperature.

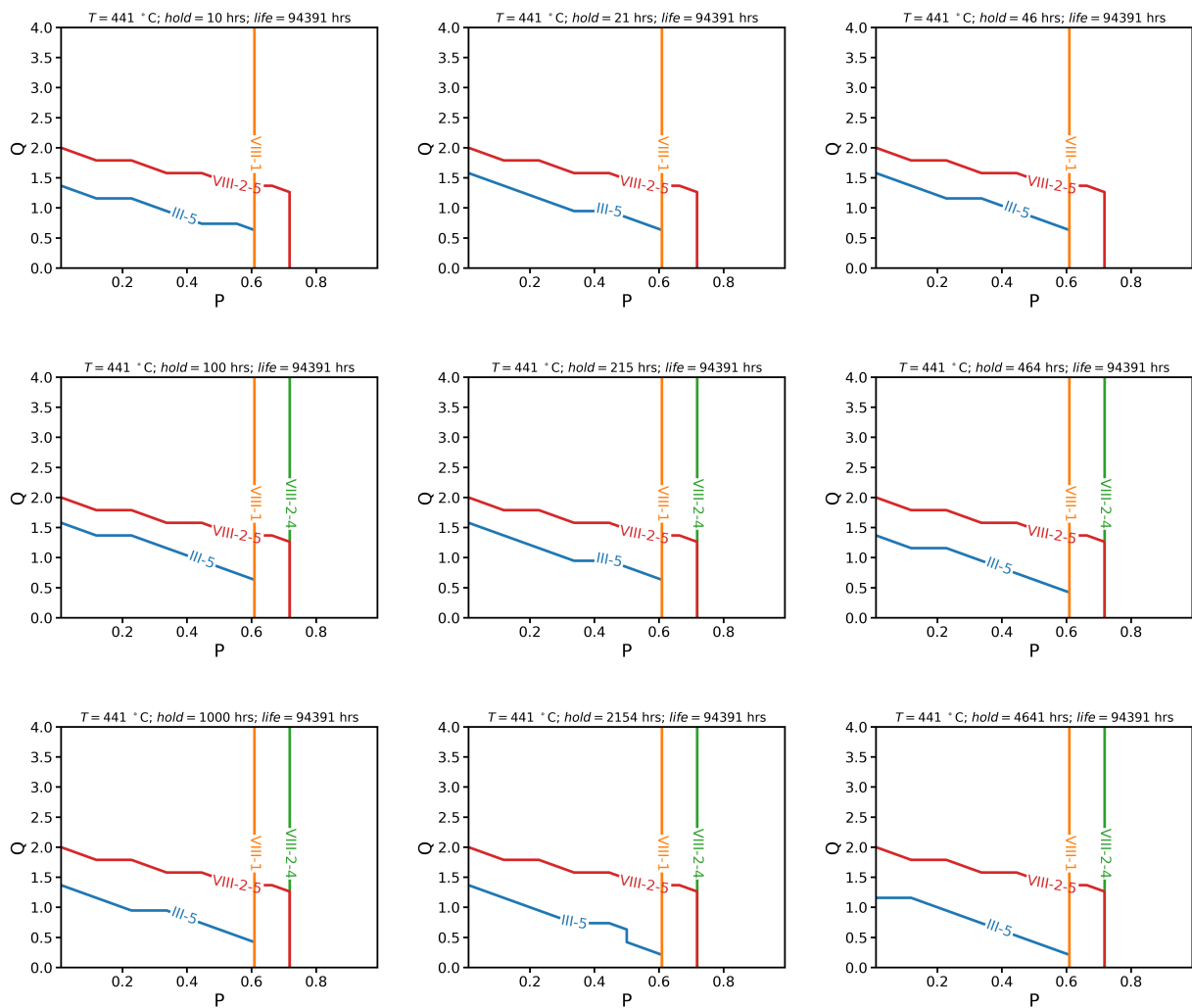


Figure 4.11. Hold time series for Grade 22 at a cooler temperature.

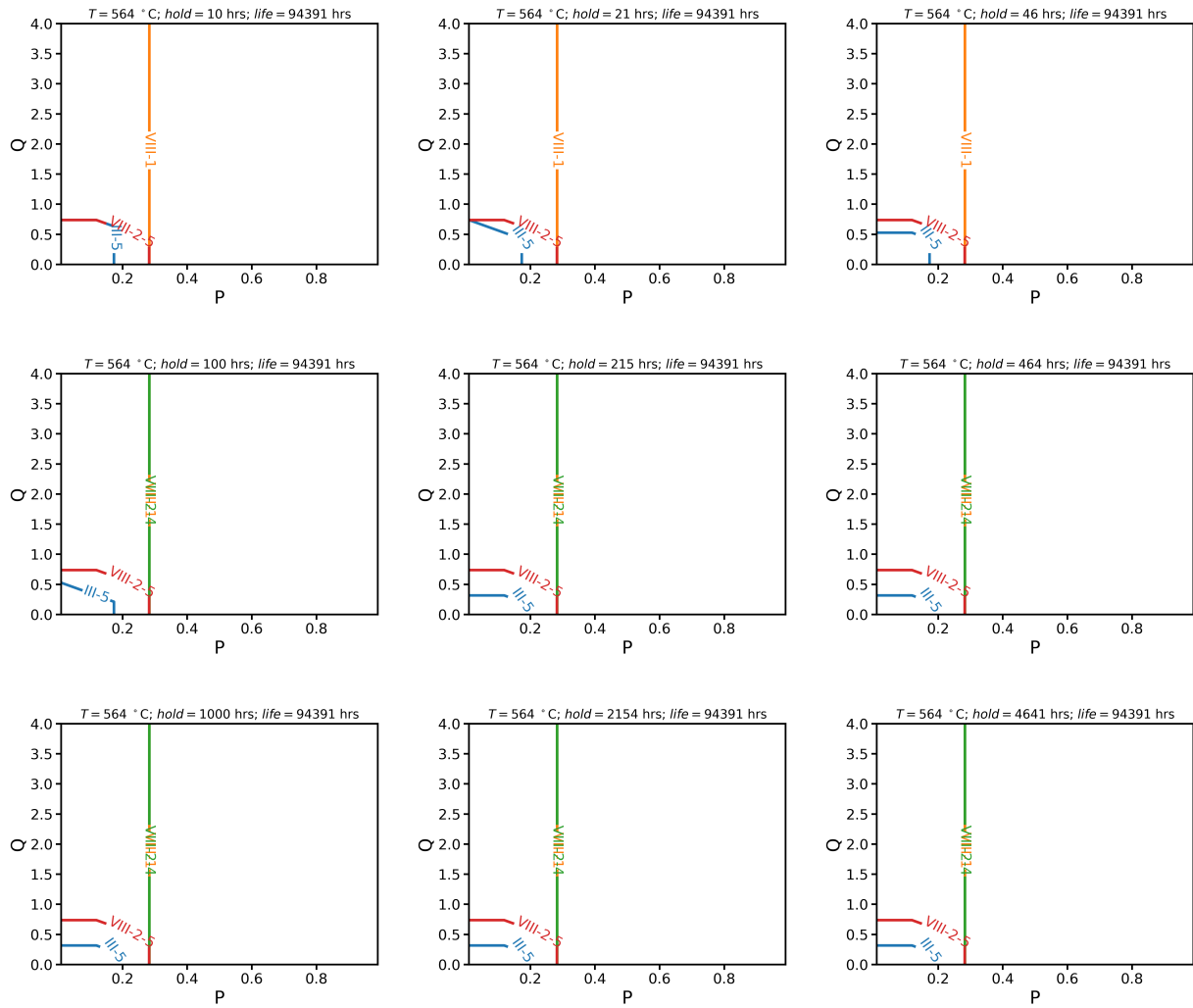


Figure 4.12. Hold time series for Grade 22 at a warmer temperature.

For the most part, these results again reinforce the conclusions drawn from the temperature study. However, this series does illustrate two new features of interest. The fatigue screening criteria for Section VIII, Division 2, Part 4 now eliminates the designs with the shortest hold times, as they experience more than 1,000 cycles over the design life. However, recall this 1,000-cycle limit cannot be properly applied to elevated temperature components, and this analysis applying this limit in lieu of the “similar operating experience” screening that would be allowed at elevated temperatures. The Section III, Division 5 feasible space both expands and contracts as a function of hold time. This represents the interplay between having more fatigue cycles, at lower hold times, and accumulating more creep damage, at longer hold times.

4.4 Overall conclusions

In general, this study reinforces conclusions that could be drawn from the analysis of the design methods in Chapter 3:

1. The Section III, Division 5 rules generally allow for lower allowable stresses than the Section VIII rules.

2. The Section VIII design by rule approaches do not cap the secondary stress, except through the fatigue screening test for Division 2. The only applicable screening criteria that applies for elevated temperature designs is screening by comparable service experience.
3. The Section VIII methods are largely insensitive to design life and cycle hold time. This mostly follows from the indefinite life approach adopted by Section VIII and the lack of creep-fatigue evaluation. However, it is notable that the design envelope barely changes even though changing the design life and hold time, in this particular problem definition, does change the number of fatigue cycles.
4. The differences between the Section VIII design envelopes and the Section III, Division 5 design envelope increase as temperature and design life increase.

Studies of this type could form the basis for a screening criterion for the use of Section VIII methods for the design of high temperature reactor components. If a design exists in the region where the Section VIII and Section III, Division 5 feasible design spaces overlap then it clearly does not matter which set of design rules the designer applies. Unfortunately, this overlap, dictated by the Section III, Division 5 design space, becomes very small at high temperatures. Based on these results, the only safe region passing both Section III and VIII rules is the triangle defined by the points $(P, Q) = (0, 0.5)$ and $(0.2, 0)$.

However, at more moderate temperatures, i.e., in the creep regime but 150 °C or more below their maximum use temperature, where these materials might more reasonably be applied in a future non-light water reactor, there is a larger region of overlap, which again could be defined by a triangle with the points $(0, 0.75)$ and $(0.3, 0)$. This criterion, more generally applied by considering the categorized primary and secondary stress calculated by elastic analysis anywhere in a component, might serve as a basis for the use of Section VIII in lieu of Section III, Division 5. While this would require some design by analysis to calculate the local stresses, it would still leave the other benefits of Section VIII. Additional work would be needed to finalize the recommended region based on additional analyses of the data collected here for this design study.

5 Preliminary conclusions

5.1 Key differences in the design rules

Chapter 3 of this report summarizes the key differences between the Section VIII and Section III, Division 5 design rules. The key conclusions from this analysis are:

- Only a subset of the Section VIII rules can be applied to high temperature design. Specifically, the rules for Section VIII, Division 2, Class 1 components do not support elevated temperature design and only the design by elastic analysis methods can be applied to components operating in the high temperature regime, subject to the gap in the current rules as written discussed in the footnote in Chapter 1.
- Section VIII bases designs on an indefinite life concept (with allowable stresses based on extrapolated 100,000-hour properties). Section III, Division 5 components have a finite design life based on time-dependent allowable stresses.
- The allowable stresses used in Section VIII design are generally higher than the allowable stresses for Section III, Division 5 at 100,000 hours.
- In current applications of Section VIII rules, the plant's remaining life and fitness for service programs are commonly employed. These programs monitor operating components and provide updated, estimated remaining lives accounting for the actual plant operating conditions and degradation of the material in service.
- None of the various Section VIII rules considered here adequately account for high temperature cyclic failure (i.e., creep-fatigue).
- The Section VIII approach to the design of weldments at high temperatures does not account for the reduced strength of weldments compared to base metal, unlike Section I, and Section III, Division 5.
- Section VIII makes a wide variety of materials available to the design, which are not available through Section III, Division 5. However, high temperature fatigue, creep-fatigue and weld strength reduction test data are not available for most of these materials.

5.2 Regions of similarity and differences in design space

Chapter 4 directly compares the design space of several sets of design rules: Section VIII, Division 1, Section VIII, Division 2, Class 2, Part 4, Section VIII, Division 2, Class 2, Part 5, and Section III, Division 5, Class A. The chapter uses the Bree cylinder as a tool to make this comparative assessment. The results of the analysis broadly match what is expected from a general consideration of the design rules:

1. The design by rule methods do not cap the secondary stress nor the number of repetitions of the load cycle.
2. The Section VIII methods do not explicitly consider design life and hold time.
3. The differences between the Section III, Division 5 and Section VIII methods are more pronounced at high temperatures and longer times.

This analysis may provide a path forward to mapping out regions of equivalency in the general design space parameterized by the Bree diagram where the use of the Section VIII rules would be acceptable simply because they provide the same design result as the Section III, Division 5 methods. At the present, however, this analysis does not apply to welded constructions.

5.3 Observations on supplementing the Section VIII rules

This section summarizes a few preliminary recommendations on supplemental design practices and material data reactor developers could use to support the use of a Section VIII component in a future advanced non-light water reactor. These recommendations only cover design rules, additional supplemental requirements may be needed for other aspects of construction.

Figure 5.1 is a flowchart illustrating the path recommended here for applying the Section VIII rules, covering items 2 to 7 in this list. Item 8 is a Section III, Division 5 alternative to the Section VIII rules. The final item related to the difference in margin in the allowable stress definition could be expanded with a statistical analysis of material strength data for a few materials.

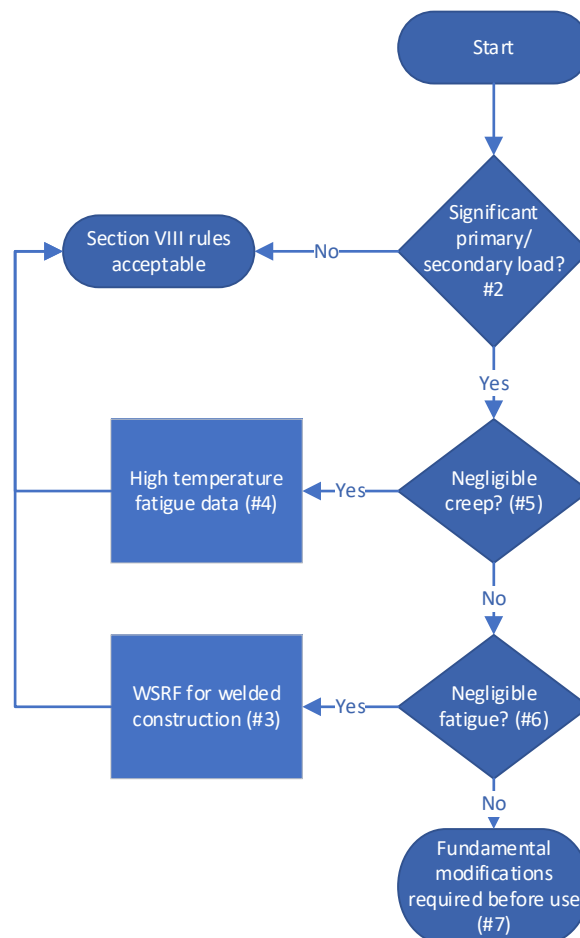


Figure 5.1 Flow chart outlining the paths for using the Section VIII rules discussed in this section.

1. The general Section VIII design philosophy provides a reduced margin in the design allowables compared to the methodology in Section III, Division 5. Failed Section VIII components are often repaired and returned to service. It may be appropriate that this difference in design philosophy be reflected, for nuclear systems, in the types of components that might be designed to Section VIII. For example, Section VIII rules might be adequate for components where failures might be tolerated without significant consequences to the rest of the reactor system or where monitoring can provide significant confidence in the integrity of a component. Additionally, it is important to note that operating plants with Section VIII components commonly employ fitness for service remaining life assessments of key components, to monitor component degradation in service and provide estimates for the actual component life to inform repair and replacement decisions.
2. The comparisons in Chapter 4 illustrate that there are regions of design space where the Section VIII and Section III, Division 5 rules overlap. In these regions, the use of Section VIII might be considered similar to the use of Section III, Division 5, as the design would pass both sets of design rules. Chapter 4 summarizes these regions in greater detail, but they are, in general, components with low primary and secondary stresses. Chapter 4 provides general guidance on specifically identifying these regions in terms of the non-dimensionalized primary and secondary stresses, P and Q .
3. A potential approach to address the reduced strength of weldments in the creep regime compared to base material, the Section VIII allowable stresses could be supplemented with a factor, perhaps absorbed into the joint efficiency factor, describing the reduced strength of the weldment in creep compared to the base material. These factors would be analogous to the Section III, Division 5 stress rupture factors or the Section I WSRF. Initial data could come from the basis underlying the factors in either Section I or Section III.
4. When negligible fatigue cannot be demonstrated, high temperature fatigue data would be needed to design Section VIII components operating in the creep regime against cyclic load, even using the current Section VIII design methods. This data currently does not exist in Section VIII, leaving only a simplified fatigue assessment as an option for high temperature components. However, the Section VIII rules do provide a method for completing the fatigue evaluation, it is only the data that is missing. As demonstrated in Chapter 4, the Section III, Division 5 strain-based high temperature fatigue design charts can be modified for use with the Section VIII fatigue design methods. This makes six materials available. For any other material beyond these six, if a designer wished to perform the analysis, the designer would need to supply high temperature fatigue data for their vessel materials. However, Chapter 3 notes that even with adequate data the Section VIII cyclic design method entirely neglects creep-fatigue interaction and so the resulting evaluation method may not be adequate for ensuring component integrity.

5. In conditions where creep is negligible, the Section VIII fatigue design rules (provided fatigue data is available) are a reasonable approach to cyclic design. Negligible creep criteria exist in Section III, Division 5, for Class A components as part of the A-3 test for ratcheting strain evaluation in Nonmandatory Appendix T and for Class B components in Subsection HCB. These criteria, or other similar criteria, could be developed for Section VIII components. If the component will operate in the negligible creep regime, a Section VIII design might function well without additional analysis, subject to the discussion on design margin below and the need for high temperature fatigue data.
6. The Section VIII rules guard against the key failure modes identified in Chapter 3 if the component operates under negligible fatigue. Under these circumstances, a Section VIII design might be adequate without additional analysis beyond that required by the Code rules for base material design, and with supplemental creep analysis of welds. Section VIII itself contains negligible fatigue criteria (the comparable service and Method A approaches discussed in Chapter 3), however, as discussed above, there may be challenges to supporting their use for elevated temperature design. More rigorous negligible fatigue criteria could be developed for high temperature construction, which could be then used to screen components that could be conservatively designed with the remainder of the existing Section VIII rules.
7. Another option is to extend the Section VIII rules to cover the key parameter ranges identified above. For high temperature cyclic loading in conditions where neither creep nor fatigue are negligible a full creep-fatigue analysis could provide quantitative insights. The rules for this type of analysis do not currently exist in the base Section VIII methods. However, Code Case 2843 provides such design methods for a Section VIII vessel, essentially copied from the Section III, Division 5, design by elastic analysis methods. It should be noted that this Code Case is limited to the Section III, Division 5 materials. The ASME has established a joint Section I/Section VIII working group on elevated temperature design to develop and eventually publish additional, simplified rules for high temperature cyclic analysis. Such rules would provide one approach for the general application of Section VIII to high temperature components in nuclear systems.
8. Section III is introducing a code case on new Class B rules that addresses some of the failure modes not covered by Section VIII rules (which are, at least approximately, similar to the current Section III, Division 5 rules). The new Class B rules will employ design by analysis methods, variable design lifetimes, and provisions for ratcheting and creep-fatigue evaluations without significant data requirements as compared with Section III, Division 5 Class A components. The approaches provided in the new Class B rules may be applicable to address limitations within Section VIII.

9. There is a myriad of differences contributing to the difference in design margin across the Section III, Division 5 and Section VIII approaches, but one clear fundamental difference is that the Section VIII allowable stresses will generally be greater than the corresponding Section III, Division 5, Class A allowable stresses in the creep regime, even when comparing the methods at a design life of 100,000 hours. Again, these differences do not necessarily imply the Section VIII component will fail prematurely. A more detailed analysis of the reliability of components design to both sets of rules, that consider probability of failure as a function of the operating life, could provide a basis for risk-informing the application of the Section VIII design rules to high temperature reactor components.

Acknowledgements

This work was sponsored by the U.S. Department of Energy, under Contract No. DE-AC02-06CH11357 with Argonne National Laboratory, managed and operated by UChicago Argonne LLC. Funding was provided by the U.S. NRC, Office of Nuclear Regulatory Research (RES) under agreement 31310024S0037. The authors express their appreciation for the support and encouragement provided by Drs. Greg Oberson and Candace de Messieres. Their insightful comments helped maintain the focus of the report to meet the regulatory and licensing needs.

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